TABLE 11-27. Electronic/Electrical Symbols (continued).

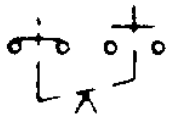
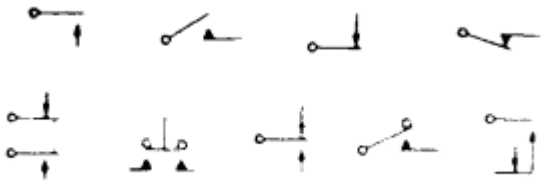
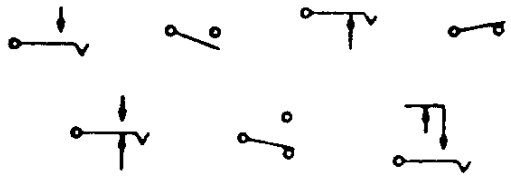
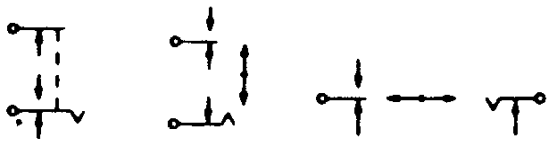
Symbol	Meaning
	Two-Circuit, Maintained, or Not Spring-Return
	Nonlocking Switching, Momentary, or Spring-Return
	Locking Switch
	Combination Locking and Nonlocking Switch
	Key-Type Switch Lever Switch

TABLE 11-27. Electronic/Electrical Symbols (continued).

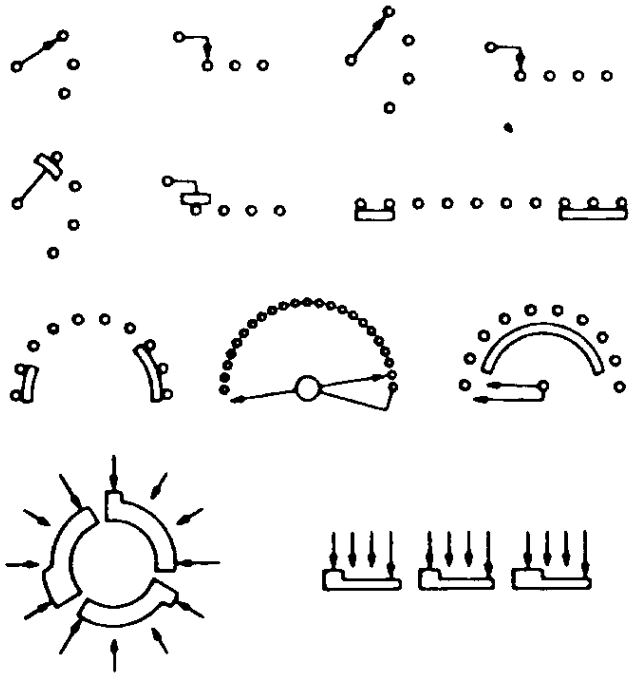
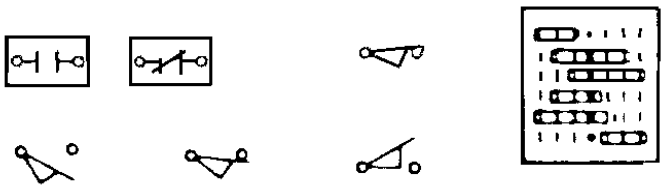
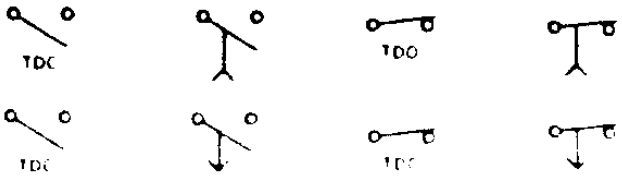
Symbol	Meaning
	Selector or Multiposition Switch
	Safety Interlock
	Limit Switch Sensitive Switch
	Switches with Time-Delay Feature
	Flow-Actuated Switch

TABLE 11-27. Electronic/Electrical Symbols (continued).

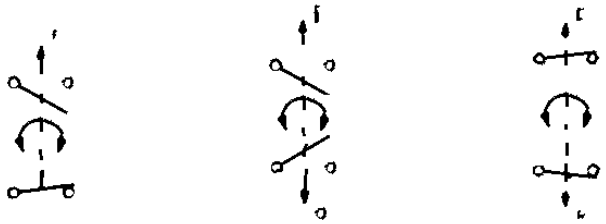
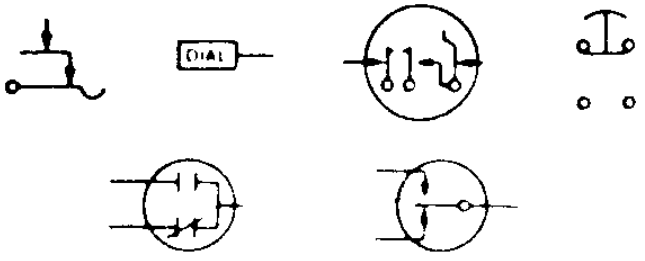
Symbol	Meaning
	Liquid-Level Actuated Switch
	Pressure- or Vacuum-Actuated Switch
	Temperature-Actuated Switch
	Thermostat
	Flasher Self-Interrupting Switch
	Foot-Operated Switch Foot Switch
	Switch Operated by Shaft Rotation and Responsive to Speed or Direction
	Switches with Specific Features

TABLE 11-27. Electronic/Electrical Symbols (continued).

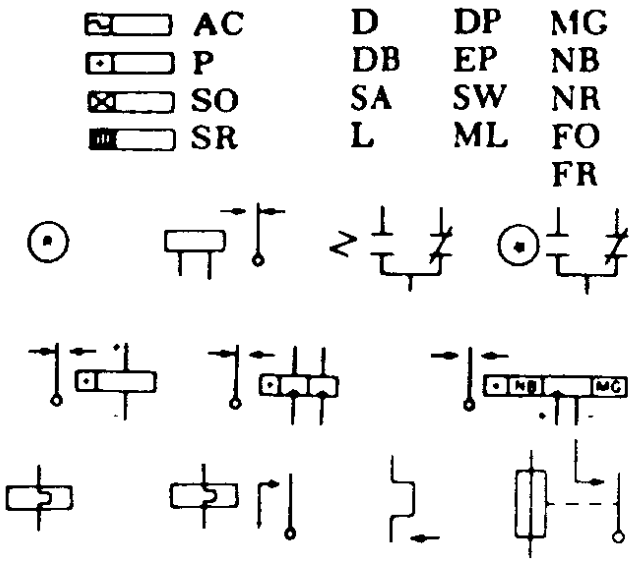
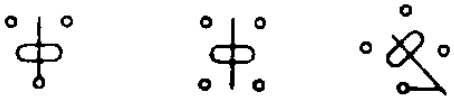
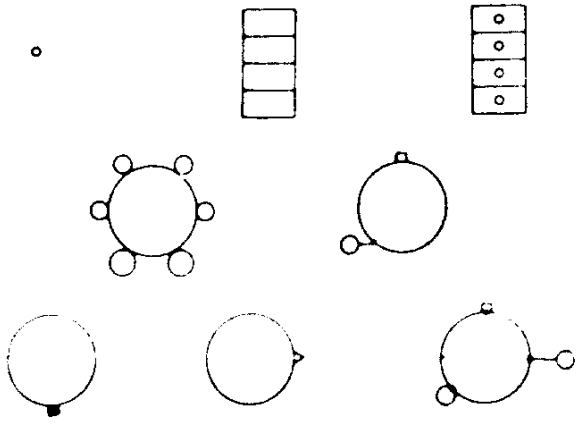
Symbol	Meaning
	Governor Speed Regulator
	Relay
	Inertia Switch
	Mercur Switch
	Terminals

TABLE 11-27. Electronic/Electrical Symbols (continued).

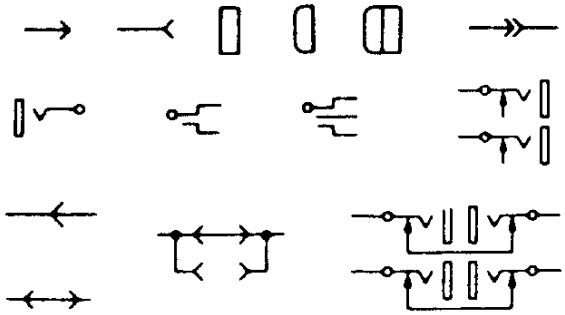
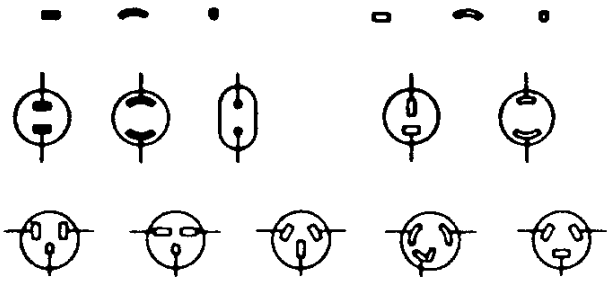
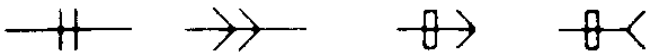
Symbol	Meaning
	Cable Termination
	Connector Disconnecting Device
	Connectors of the Type Commonly Used for Power-Supply Purposes
	Test Blocks
	Coaxial Connector
	Waveguide Flanges Waveguide Junction

TABLE 11-27. Electronic/Electrical Symbols (continued).

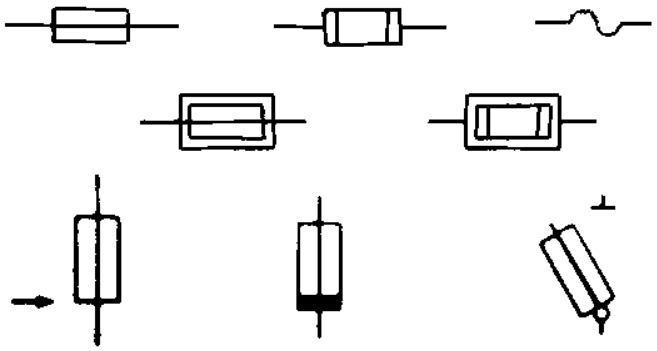
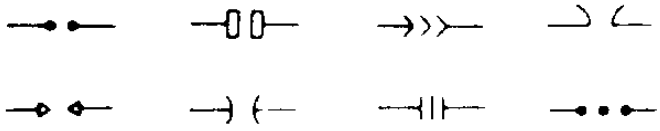
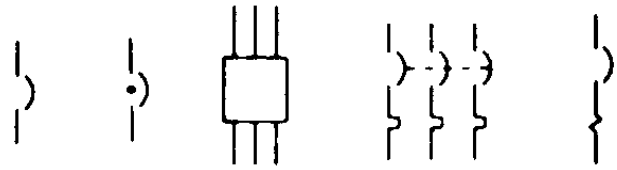
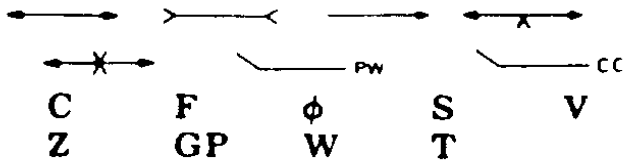
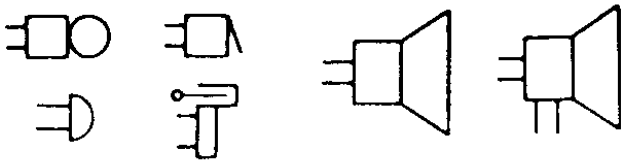
Symbol	Meaning
	Fuse
	Lightning Arrester Arrester Gap
	Circuit Breaker
	Protective Relay
	Audible-Signaling Device
	Microphone

TABLE 11-27. Electronic/Electrical Symbols (continued).

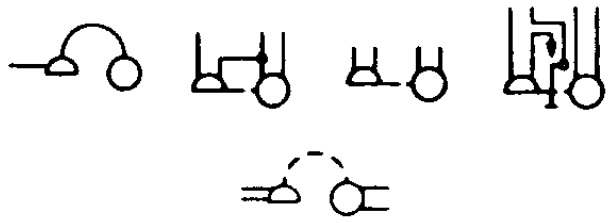
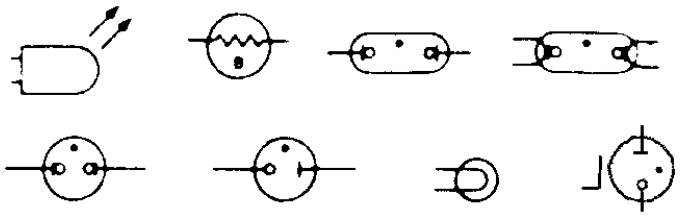
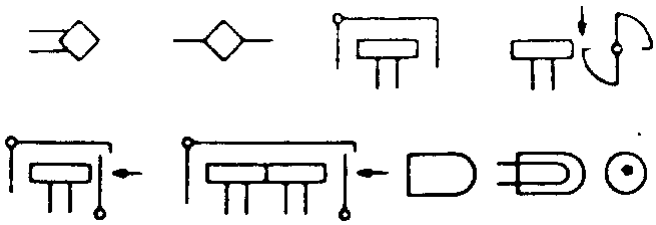
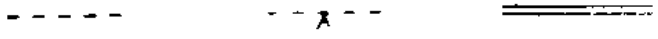
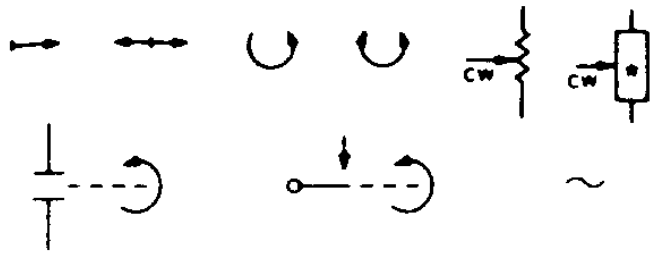
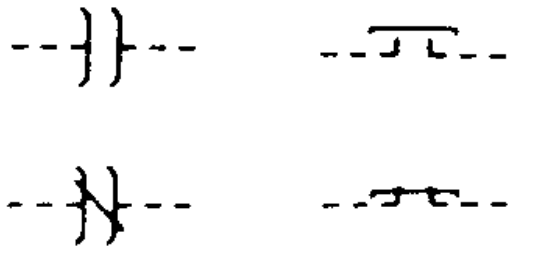
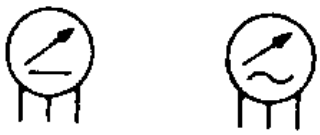
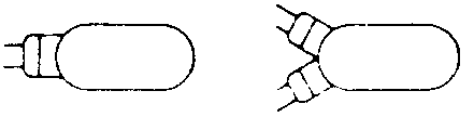
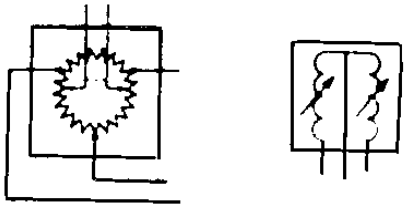
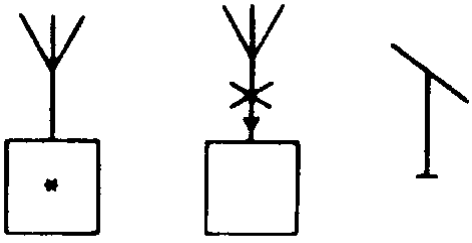
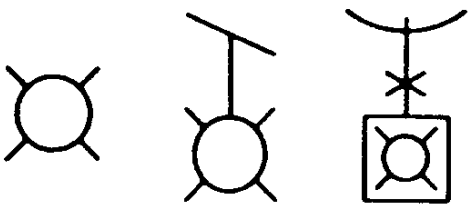
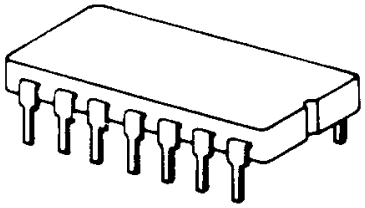
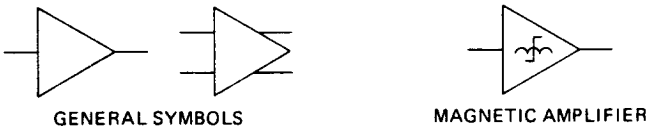
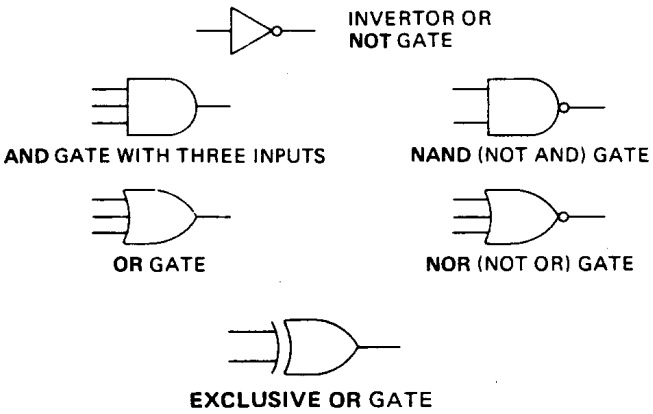
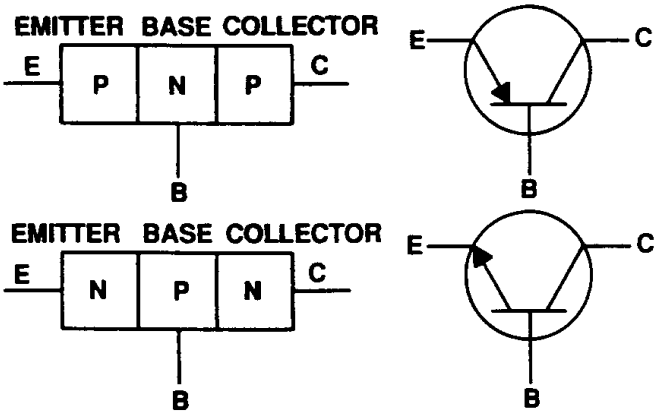
Symbol	Meaning
	Handset Operator's Set
	Lamp
	Visual-Signaling Device
	Mechanical Connection Mechanical Interlock
	Mechanical Motion
	Clutch Brake

TABLE 11-27. Electronic/Electrical Symbols (continued).

Symbol	Meaning
	Manual Control
	Gyro Gyroscope Gyrocompass
	Position Indicator
	Fire Extinguisher Actuator Head
	Position Transmitter
	Radio Station
	Space Station

	Integrated Circuit
	Amplifiers
	Logic Gates
	Diode
	Transistor Symbols

11-273.—11-283. [RESERVED.]

CHAPTER 12. AIRCRAFT AVIONICS SYSTEMS

SECTION 1. AVIONICS EQUIPMENT MAINTENANCE

12-1. GENERAL. There are several methods of ground checking avionics systems.

a. Visual Check. Check for physical condition and safety of equipment and components.

b. Operation Check. This check is performed primary by the pilot, but may also be performed by the mechanics after annual and 100-hour inspections. The aircraft flight manual, the Airman's Information Manual (AIM), and the manufacturer's information are used as a reference when performing the check.

c. Functional Test. This is performed by qualified mechanics and repair stations to check the calibration and accuracy of the avionics with the use of test equipment while they are still on the aircraft, such as the transponder and the static checks. The equipment manufacturer's manuals and procedures are used as a reference.

d. Bench Test. When using this method the unit or instrument is removed from the aircraft and inspected, repaired, and calibrated as required.

e. Electromagnetic Interference (EMI). For EMI tests, refer to chapter 11 paragraph 11-107 of this AC.

12-2. HANDLING OF COMPONENTS. Any unit containing electronic components such as transistors, diodes, integrated circuits, proms, roms, and memory devices should be protected from excessive shocks. Excessive shock can cause internal failures in an of these components. Most electronic devices are subject to damage by electrostatic discharges (ESD).

CAUTION: To prevent damage due to excessive electrostatic discharge, proper gloves, finger cots, or grounding bracelets should be used. Observe the standard procedures for handling equipment containing electrostatic sensitive devices or assemblies in accordance with the recommendations and procedures set forth in the maintenance instructions set forth by the equipment manufacturers.

12-3.—12-7. [RESERVED.]

SECTION 2. GROUND OPERATIONAL CHECKS FOR AVIONICS EQUIPMENT (ELECTRICAL)

12-8. GENERAL. When the operating or airworthiness regulations require a system to perform its intended function, the use of the Technical Standard Order (TSO) equipment or the submission of data substantiating the equipment performance is strongly recommended. An operation check of avionics is the responsibility of the pilot in command. However, it is recommended that after replacement of equipment during 100 hour or annual inspections, an operational check of avionics equipment be performed. The accomplishments of these checks must be done in accordance with the recommendations and procedures set forth in the aircraft's flight manual instructions published by the avionics equipment manufacturers.

12-9. INSPECTION OF AVIONICS SYSTEMS.

a. The inspection shall include the following:

(1) Inspect the condition and security of equipment including the proper security of wiring bundles.

(2) Check for indications of overheating of the equipment and associated wiring.

(3) Check for poor electrical bonding. The bonding requirements are specified by equipment manufacturers. Installation cabling should be kept as short as possible, except for antenna cables which are usually precut or have a specific length called out at installation. Proper bonding on the order of 0.003 ohms is very important to the performance of avionics equipment.

(4) Check to assure that the radios and instruments are secured to the instrument panel.

(5) Check that all avionics are free of dust, dirt, lint, or any other airborne contaminates. If there is a forced air cooling system, it must be inspected for proper operation. Equipment ventilation openings must not be obstructed.

(6) Check the microphone headset plugs and connectors and all switches and controls for condition and operation. Check all avionics instruments for placards. Check lightening, annunciator lights, and cockpit interphone for proper operation.

(7) The circuit breaker panel must be inspected for the presence of placarding for each circuit breaker installed.

(8) Check the electrical circuit switches, especially the spring-load type for proper operation. An internal failure in this type of switch may allow the switch to remain closed even though the toggle or button returns to the OFF position. During inspection, attention must be given to the possibility that improper switch substitution may have been made.

b. Inspect antennas for:

(1) broken or missing antenna insulators

(2) lead through insulators

(3) springs

(4) safety wires

(5) cracked antenna housing

(6) missing or poor sealant at base of antenna

(7) correct installation

(8) signs of corrosion, and

(9) the condition of paint/bonding and grounding.

(10) Check the bonding of each antenna from mounting base to the aircraft skin. Tolerance: .1 ohm, maximum.

(a) Test Equipment:

1 1502B Metallic Time Domain Reflectometer or equivalent.

2 Thruline Wattmeter.

(b) Perform the antenna evaluation check using the domain reflectometer to determine the condition of the antenna and coax cables. Refer to manufacturer's maintenance procedures.

(c) Use thruline wattmeter as needed for addition evaluation. Refer to manufacturer's maintenance procedures. Check for the following:

1 Resistance.

2 Shorts.

3 Opens.

c. Inspect the static dischargers/wicks for:

(1) physical security of mounting attachments, wear or abrasion of wicks, missing wicks, etc.,

(2) assurance that one inch of the inner braid of flexible vinyl cover wicks extends beyond the vinyl covering,

(3) assurance that all dischargers are present and securely mounted to their base,

(4) assurance that all bases are securely bonded to skin of aircraft in order to prevent the existence in voltage level differences between two surfaces,

(5) signs of excessive erosion or deterioration of discharger tip,

(6) lighting damage as evidenced by pitting of the metal base, and

(7) megohm value of static wick itself as per manufacturer's instructions. It should not be open.

d. Subsequent inspection must be made after a maintenance action on a transponder. Refer to Title 14 of the Code of Federal Regulations (14 CFR) part 91, sections 91.411 and 91.413.

e. Inspection of the emergency locator transmitter operation, condition and date of the battery.

f. Perform a function check of the radio by transmitting a request for a radio check. Perform a function check on navigation equipment by moving the omni bearing selection (OBS) and noting the needle swing and the TO/FROM flag movement.

12-10. COMMUNICATION SYSTEMS. Ground operation of communication systems in aircraft may be accomplished in accordance with the procedures appropriate for the airport and area in which the test is made and the manufacturer's manuals and procedures. Check system(s) for side tone, clarity of

transmission, squelch, operations using head phones, speaker(s), and hand microphone. If a receiver or transmitter is found to be defective, it should be removed from the aircraft and repaired.

12-11. VHF OMNI-DIRECTIONAL RANGE (VOR). A VOR operates within the 108.0 to 111.85 MHz, and 112.0 to 117.95 MHz frequency bands. The display usually consists of a deviation indicator and a TO/FROM indicator. The controls consist of a frequency selector for selecting the ground station and an OBS, which is used for course selection. An ON/OFF flag is used to determine adequate field strength and presence of a valid signal. There are numerous configurations when integrated into flight directors and/or when using a slaved compass system, which uses an additional indicator that points continually to the selected omni station regardless of OBS selection. In order to determine the accuracy specified in a functional check, a ground test set must be used in accordance with the manufacturer's specifications. For the purpose of this inspection/maintenance activity, the following operational check can be accomplished to determine if the equipment has the accuracy required for operation in instrument flight rules (IFR) environment. Verify audio identification, OBS operation, flag operation, radio magnetic indicator (RMI) interface, and applicable navigation (NAV) switching functions. The operational check is also published in the AIM, section 1-1-4. This check is required by 14 CFR part 91, section 91.171 before instrument flight operations.

12-12. DISTANCE MEASURING EQUIPMENT (DME). The operation of DME consists of paired pulses at a specific spacing, sent out from the aircraft (this is what is called interrogation), and are received by the ground station, which then responds with paired pulses at the specific spacing sent by the aircraft, but at a different frequency. The aircraft unit measures the time it takes to transmit and then receive the signal, which then is translated into distance. DME operates on frequencies from 962 MHz to 1213 MHz. Because of the curvature of earth, this line-of-sight signal is reliable up to 199 nautical mile (NM) at the high end of the controlled airspace with an accuracy of 1/2 mile or 3 percent of the distance. DME inspection/maintenance on the aircraft is most commonly limited to a visual check of the installation, and if there have been previously reported problems, the antenna must be inspected for proper bonding and the absence of corrosion, both on the mounting surface, as well as the coax connector. Accuracy can be determined by evaluating performance during flight operations, as well as with ground test equipment. If a discrepancy is reported and corrected, it is good practice to make the accuracy determination before instrument flight. Tune the DME to a local station, or use the proper ground test equipment to check audio identification, and DME hold function verify correct display operation.

12-13. AUTOMATIC DIRECTION FINDER (ADF). The ADF receivers are primarily designed to receive nondirectional beacons (NDB) in the 19 to 535 kHz amplitude modulation (AM) broadcast low band. The receivers will also operate in the commercial AM band. The ADF display pointer will indicate the relative bearing to a selected AM band transmitter that is in range. An ADF system must be checked by tuning to an adequate NDB or commercial AM station. Verify proper bearing to station, audio identification and tone/beat frequency oscillator (BFO), correct operation in closed circuit (LOOP) and sense modes. Note the orientation of the selected station to the aircraft using an appropriate chart. Observe the ADF relative bearing reading, and compare to the chart. Slew the needle and observe how fast (or slowly) it returns to the reading. ADF performance may

be degraded by lightning activity, airframe charging, ignition noise and atmospheric phenomena.

12-14. INSTRUMENT LANDING SYSTEMS (ILS). The ILS consist of several components, such as the localizer, glide slope, marker beacon, radio altimeter, and DME. Localizer and glide slope receivers and marker beacons will be discussed in this section.

a. Localizer receiver operates on one of 40 ILS channels within the frequency range of 108.10 to 111.95 MHz (odd tens). These signals provide course guidance to the pilot to the runway centerline through the lateral displacement of the VOR/localizer (LOC) deviation indicator. The ground transmitter is sighted at the far end of the runway and provides a valid signal from a distance of 18 NM from the transmitter. The indication gives a full fly left/right deviation of 700 feet at the runway threshold. Identification of the transmitter is in International Morse Code and consists of a three letter identifier preceded by the Morse Code letter I (two dots). The localizer function is usually integral with the VOR system, and when maintenance is performed on the VOR unit, the localizer is also included. The accuracy of the system can be effectively evaluated through normal flight operations if evaluated during visual meteorological conditions. Any determination of airworthiness after reinstallation before instrument flight must be accomplished with ground test equipment.

b. The glide slope receiver operates on one of 40 channels within the frequency range 329.15 MHz, to 335.00 MHz. The glide slope transmitter is located between 750 feet and 1250 feet from the approach end of the runway and offset 250 to 650 feet. In the absence of questionable performance, periodic functional flight checks of the glide slope system would be an acceptable way to ensure continued system performance. The functional flight test must be conducted under visual flight rules (VFR) conditions. A failed or misleading system must be serviced by an appropriately-rated repair station. Ground test equipment can be used to verify glide slope operation.

c. Localizer/Glide Slope (LOC/GS) may have self test function, otherwise the proper ground test equipment must be used. Refer to manufacturer's or aircraft instruction manual.

12-15. MARKER BEACON. Marker beacon receivers operate at 75 MHz and sense the audio signature of each of the three types of beacons. The marker beacon receiver is not tunable. The blue outer marker light illuminates when the receiver acquires a 75 MHz signal modulated with 400 Hz, an amber middle marker light for a 75 MHz signal modulated with 1300 Hz and, a white inner marker light for a 75 MHz signal modulated with 3000 Hz. The marker beacon system must be operationally evaluated in VFR when an ILS runway is available. The receiver sensitivity switch must be placed in LOW SENSE (the normal setting). Marker audio must be adequate. Ground test equipment must be used to verify marker beacon operation. Marker beacon with self test feature, verify lamps, audio and lamp dimming.

12-16. LONG RANGE NAVIGATION (LORAN). The LORAN has been an effective alternative to Rho/Theta R-Nav systems. Hyperbolic systems require waypoint designation in terms of latitude and longitude, unlike original R-Nav (distance navigation) systems, which define waypoints in terms of distance (Rho) and angle (Theta) from established VOR or Tacan facilities. Accuracy is better than the VOR/Tacan system but LORAN is more prone to problems with precipitation static. Proper bonding of aircraft structure and the use of high-quality static wicks will not only produce improved LORAN system performance, but can also benefit the very high frequency

(VHF) navigation and communications systems. This system has an automatic test equipment (ATE).

NOTE: Aircraft must be outside of hangar for LORAN to operate.

Normally self test check units, verification of position, and loading of flight plan will verify operation verification of proper flight manual supplements and operating handbooks on board, and proper software status can also be verified.

12-17. GLOBAL POSITIONING SYSTEM (GPS). The GPS is at the forefront of present generation navigation systems. This space-based navigation system is based on a 24-satellite system and is highly accurate (within 100 meters) for establishing position. The system is unaffected by weather and provides a world-wide common grid reference system. Database updating and antenna maintenance are of primary concern to the GPS user.

NOTE: Aircraft must be outside of hangar for ground test of GPS.

12-18. AUTOPILOT SYSTEMS. Automatic Flight Control Systems (AFCS) are the most efficient managers of aircraft performance and control. There are three kinds of autopilot; two axes, three axes, and three axes with coupled approach capability. Attention must be given to the disconnect switch operation, aural and visual alerts of automatic and intentional autopilot disconnects, override forces and mode annunciation, servo operation, rigging and bridle cable tension, and condition. In all cases the manufacturer's inspection and maintenance instructions must be followed.

12-19. ALTIMETERS. Aircraft conducting operations in controlled airspace under instrument flight rule (IFR) are required to have their static system(s) and each altimeter instrument inspected and tested within the previous 24 calendar months. Frequent functional checks of all altimeters and automatic pressure altitude reporting systems are recommended.

a. The tests required must be performed by:

(1) The manufacturer of the aircraft on which the tests and inspection are to be performed.

(2) A certificated repair station properly equipped to perform those functions and holding:

(a) An instrument rating Class I.

(b) A limited instrument rating appropriate to the make and model of appliance to be tested.

(c) A certified/qualified mechanic with an airframe rating(static system tests and inspections only). Any adjustments shall be accomplished only by an instrument shop certified/qualified person using proper test equipment and adequate reference to the manufacturer's maintenance manuals. The altimeter correlation adjustment shall not be adjusted in the field. Changing this adjustment will nullify the correspondence between the basic test equipment calibration standards and the altimeter. It will also nullify correspondence between the encoding altimeter and its encoding digitizer or the associated blind encoder.

b. Examine the altimeter face for evidence of needle scrapes or other damage. Check smoothness of operation, with particular attention to altimeter performance during decent.

c. Contact an appropriate air traffic facility for the pressure altitude displayed to the controller from your aircraft. Correct the reported altitude as needed, and compare to the

reading on the altimeter instrument. The difference must not exceed 125 feet.

12-20. TRANSPONDERS. There are three modes (types) of transponders that can be used on various aircraft. Mode A provides a (non altitude-reporting) four-digit coded reply; Mode C provides a code reply identical to Mode A with an altitude-reporting signal; and Mode S has the same capabilities as Mode A and Mode C and responds to traffic alert and collision avoidance system (TCAS)-Equipped Aircraft.

a. Ground ramp equipment must be used to demonstrate proper operation. Enough codes must be selected so that each switchposition is checked at least once. Low and high sensitivity operation must be checked. Identification operation must be checked. Altitude reporting mode must be demonstrated. Demonstrate that the transponder system does not interfere with other systems aboard the aircraft, and that other equipment does not interfere with transponder operation. Special consideration must be given to other pulse equipment, such as DME and weather radar.

b. All transponders must be tested every 24-calendar months, or during an annual inspection, if requested by the owner. The test must be conducted by an authorized avionics repair facility.

12-21. EMERGENCY LOCATOR TRANS- MITTERS (ELT). The ELT must be evaluated in accordance with TSO-C91a, TSO-C126 for 406 MHz ELT's, or later TSO's issued for ELT's. ELT installations must be examined for potential operational problems at least once a year (section 91.207(d)). There have been numerous instances of interaction between ELT and other VHF installations. Antenna location should be as far as possible from other antennas to prevent efficiency losses. Check ELT antenna installations in close proximity to other VHF antennas for suspected interference. Antenna patterns of previously installed VHF antennas could be measured after an ELT installation. Tests should be conducted during the first 5 minutes after any hour. If operational tests must be made outside of this time frame, they should be coordinated with the nearest FAA Control Tower or FSS. Tests should be no longer than three audible sweeps.

12-22. INSPECTION OF ELT. An inspection of the following must be accomplished by a properly certified person or repair station within 12-calendar months after the last inspection:

a. Proper Installation.

(1) Remove all interconnections to the ELT unit and ELT antenna. Visually inspect and confirm proper seating of all connector pins. Special attention should be given to coaxial center conductor pins, which are prone to retracting into the connector housing.

(2) Remove the ELT from the mount and inspect the mounting hardware for proper installation and security.

(3) Reinstall the ELT into its mount and verify the proper direction for crash activation. Reconnect all cables. They should have some slack at each end and should be properly secured to the airplane structure for support and protection.

b. Battery Corrosion. Gain access to the ELT battery and inspect. No corrosion should be detectable. Verify the ELT battery is approved and check its expiration date.

c. Operation of the Controls and Crash Sensor. Activate the ELT using an applied force. Consult the ELT manufacturer's instructions before activation. The direction for mounting and force activation is indicated on

the ELT. A TSO-C91 ELT can be activated by using a quick rap with the palm. A TSO-C91a ELT can be activated by using a rapid forward (throwing) motion coupled by a rapid reversing action. Verify that the ELT can be activated using a watt meter, the airplane's VHF radio communications receiver tuned to 121.5 MHz, or other means (see NOTE 1). Insure that the "G" switch has been reset if applicable.

d. For a Sufficient Signal Radiated From its Antenna. Activate the ELT using the ON or ELT TEST switch. A low-quality AM broadcast radio receiver should be used to determine if energy is being transmitted from the antenna. When the antenna of the AM broadcast radio receiver (tuning dial on any setting) is held about 6 inches from the activated ELT antenna, the ELT aural tone will be heard (see NOTE 2 and 3).

e. Verify that All Switches are Properly Labeled and Positioned.

f. Record the Inspection. Record the inspection in the aircraft maintenance records according to 14 CFR part 43, section 43.9. We suggest the following:

I inspected the Make/Model _____ ELT system in this aircraft according to applicable Aircraft and ELT manufacturer's instructions and applicable FAA guidance and found that it meets the requirements of section 91.207(d).

Signed: _____
Certificate No. _____
Date: _____

NOTE 1: This is not a measured check; it only indicates that the G-switch is working.

NOTE 2: This is not a measured check; but it does provide confidence that the antenna is radiating with sufficient power to aid search and rescue. The signal may be weak even if it is picked up by an aircraft VHF receiver located at a considerable distance from the radiating ELT. Therefore, this check does not check the integrity of the ELT system or provide the same level of confidence as does the AM radio check.

NOTE 3: Because the ELT radiates on the emergency frequency, the Federal Communications Commission allows these tests only to be conducted within the first five minutes after any hour and is limited in three sweeps of the transmitter audio modulation.

12-23. FLIGHT DATA RECORDER. The flight data recorder is housed in a crush-proof container located near the tail section of the aircraft. The tape unit is fire resistant, and contains a radio transmitter to help crash investigators locate the unit under water. Inspection/Operational checks include:

a. Check special sticker on front of the flight data recorder for the date of the next tape replacement, if applicable.

b. Remove recorder magazine and inspect tape for the following:

(1) broken or torn tape,

(2) proper feed of tape, and

(3) all scribes were recording properly for approximately the last hour of flight.

c. Conditions for tape replacement (as applicable):

(1) There is less than 20 hours remaining in the magazine as read on the *tape remaining* indicator.

(2) Tape has run out.

(3) Broken tape.

(4) After hard landings and severe air turbulence have been encountered as reported by the pilots. After the same tape has been in use 1 year (12 months), it must be replaced.

(5) Ensure that a correlation test has been performed and then recorded in the aircraft records.

d. Refer to the specific equipment manufacturer's manuals and procedures.

e. The state-of-the art Solid-State Flight Data Recorder (SSFDR) is a highly flexible model able to support a wide variety of aeronautical radio, incorporated (ARINC) configurations. It has a Built-In Test Equipment (BITE) that establishes and monitors the mission fitness of the hardware. BITE performs verification after storage (read after write) of flight data and status condition of the memory. These recorders have an underwater acoustic beacon mounted on its front panel which must be returned to their respective manufacturer's for battery servicing. For maintenance information refer to the equipment or aircraft manufacture's maintenance instruction manual.

12-24. COCKPIT VOICE RECORDERS (CVR). CVR's are very similar to flight data recorders. They look nearly identical and operate in almost the same way. CVR's monitors the last 30 minutes of flight deck conversations and radio communications. The flight deck conversations are recorded via the microphone monitor panel located on the flight deck. This panel is also used to test the system and erase the tape, if so desired. Before operating the erase CVR mode, consult the operational manual of the manufacturer for the CVR.

a. Playback is possible only after the recorder is removed from the aircraft.

b. Refer to the specific equipment manufacturer's manuals and procedures.

c. The Solid State Cockpit Voice Recorder system is composed of three essential components a solid state recorder, a control unit (remote mic amplifier), and an area microphone. Also installed on one end of the recorder is an Under water Locator Beacon (ULB). The recorder accepts four separate audio inputs: pilot, copilot, public address/third crew member, and cockpit area microphone and where applicable, rotor speed input and flight data recorder synchronization tone input. For maintenance information refer to the equipment manufacturer's maintenance manual.

12-25. WEATHER RADAR. Ground performance shall include antenna rotation, tilt, indicator brilliance, scan rotation, and indication of received echoes. It must be determined that no objectionable interference from other electrical/electronic equipment appears on the radar indicator, and that the radar system does not interfere with the operation of any of the aircraft's communications or navigation systems.

CAUTION: Do not turn radar on within 15 feet of ground personnel, or containers holding flammable or explosive materials. The radar should never operate during fueling operations. Do not operate radar system when beam may intercept larger metallic objects closer than 150 feet, as crystal damage might occur. Do not operate radar when cooling fans are inoperative. Refer to the specific Ra

dar System equipment manufacturer's manuals and procedures.

12-26. RADOME INSPECTION. Inspection of aircraft having weather radar installations should include a visual check of the radome surface for signs of surface damage, holes, cracks, chipping, and peeling of paint, etc. Attach fittings and fastenings, neoprene erosion caps, and lightening strips, when installed, should also be inspected.

12-27. DATA BUS. Data Buses provide the physical and functional partitioning needed to enable different companies to design different avionics boxes to be able to communicate information to each other. It defines the framework for system(s) intergration. There are several types of data bus analyzers used to receive and review transmitted data or to transmit data to a bus user. Before using an analyzer, make sure that the bus language is compatible with the bus analyzer. For further information refer to ARINC specifications such as 429 Digital Information Transfer System, Mark 33 which offers simple and affordable answers to data communications on aircraft.

12-28. ELECTRIC COMPATABLITY. When replacing an instrument with one which provides additional functions or when adding new instruments, check the following electrical (where applicable) for compatibility:

a. Voltage (AC/DC).
b. Voltage polarity (DC).
c. Voltage phase (s) (AC).
d. Frequency (AC).
e. Grounding (AC/DC).
f. System impedance matching.
g. Compatibility with system to which connected.

12-29.—12-36. [RESERVED.]

SECTION 3. GROUND OPERATIONAL CHECKS FOR AVIONICS EQUIPMENT (NON ELECTRICAL)

12-37. COMPASS SWING must be performed whenever any ferrous component of the system (i.e. flux valve compensator, or Standby Compass) is installed, removed, repaired, or a new compass is installed. The magnetic compass can be checked for accuracy by using a compass rose located on an airport. The compass swing is normally effected by placing the aircraft on various magnetic headings and comparing the deviations with those on the deviation cards. Refer to CFR14, 23.1327, 14 CFR 23.1547, and the equipment or aircraft manufacturer's manual.

a. A compass swing must be performed on the following occasions:

(1) When the accuracy of the compass is suspected.

(2) After any cockpit modification or major replacement involving ferrous metal.

(3) Whenever a compass has been subjected to a shock; for example, after a hard landing or turbulence.

(4) After aircraft has passed through a severe electrical storm.

(5) After lighting strike.

(6) Whenever a change is made to the electrical system.

(7) Whenever a change of cargo is likely to affect the compass.

(8) When an aircraft operation is changed to a different geographic location with a major change in magnetic deviation. (e.g., from Miami, Florida to Fairbanks, Alaska.)

(9) After aircraft has been parked on one heading for over a year.

(10) When flux valves are replaced.

b. Precautions.

(1) The magnetic compass must be checked for accuracy in a location free of steel structures, underground pipes or cables, or equipment that produces magnetic fields.

(2) Personnel engaged in the compensation of the compass shall remove all magnetic or ferrous material from their possession.

(3) Use only nonmagnetic tools when adjusting the compass.

(4) Position the aircraft at least 100 yards from any metal object.

(5) All equipment in the aircraft having any magnetic effect on the compass must be secured in the position occupied in normal flight.

c. Compass Swing Procedures.

(1) Have the aircraft taxied to the NORTH (0°) radial on the Compass Rose. Use a hairline sight compass (a reverse reading compass with a gun sight arrangement mounted on top of it) to place the aircraft in the general vicinity. With the aircraft facing North and the person in the cockpit running the engine(s) at 1000 rpm, a mechanic, standing approximately 30 feet in front of the aircraft and facing South, “shoots” or aligns the master compass with the aircraft center line. Using hand signals, the mechanic signals the person in the cockpit to make additional adjustments to align the aircraft with the master

compass. Once aligned on the heading, the person in the cockpit runs the engine(s) to approximately 1,700 rpm to duplicate the aircraft's magnetic field and then the person reads the compass.

NOTE: (1) For conventional gear aircraft, the mechanic will have to position the magnetic compass in the straight and level position or mount the tail of the aircraft on a moveable dolly to simulate a straight and level cruise configuration. (2) Remember the hairline sight compass is only intended to be used as a general piece of test equipment.

(2) If the aircraft compass is not in alignment with the magnetic North of the master compass, correct the error by making small adjustments to the North-South brass adjustment screw with a nonmetallic screw driver (made out of brass stock, or stainless steel welding rod). Adjust the N-S compensator screw until the compass reads North (0°). Turn the aircraft until it is aligned with the East-West, pointing East. Adjust the E-W compensator screw until it reads 90°. Continue by turning the aircraft South 180° and adjust the N-S screw to remove one-half of the South's heading error. This will throw the North off, but the total North-South should be divided equally between the two headings. Turn the aircraft until it is heading West 270°, and adjust the E-W screw on the compensator to remove one-half of the West error. This should divide equally the total E-W error. The engine(s) should be running.

(3) With the aircraft heading West, start your calibration card here and record the magnetic heading of 270° and the compass reading with the avionics/electrical systems on then off. Turn the aircraft to align with each of the lines on the compass rose and record the com pass reading every 30°. There should be not more than a plus or minus 10° difference between any of the compass' heading and the magnetic heading of the aircraft.

(4) If the compass cannot be adjusted to meet the requirements, install another one.

NOTE: A common error that affects the compass' accuracy is the mounting of a compass or instruments on or in the instrument panel using steel machine screws/nuts rather than brass hardware, magnetized control yoke, structural tubing, and improperly routed electrical wiring, which can cause unreasonable compass error.

(5) If the aircraft has an electrical system, two complete compass checks should be performed, one with minimum electrical equipment operating and the other with all electrical accessories on (e.g. radios, navigation, radar, and lights). If the compass readings are not identical, the mechanic should make up two separate compass correction cards, one with all the equipment on and one with the equipment off.

(6) When the compass is satisfactorily swung, fill out the calibration card properly and put it in the holder in full view for the pilot's reference.

d. Standby (wet) Compass. Adjustment and compensation of the Standby Compass may also be accomplished by using the "compass swing" method.

12-38. PNEUMATIC GYROS.

a. Venturi Systems. The early gyro instruments were all operated by air flowing out of a jet over buckets cut into the periphery of the gyro rotor. A venturi was mounted on the outside of the aircraft to produce a low pressure, or vacuum, which evacuated the instrument case, and air flowed into the instrument through a paper filter and then through a nozzle onto the rotor.

(1) Venturi systems have the advantage of being extremely simple and requiring no power from the engine, nor from any of the other aircraft systems; but they do have the disadvantage of being susceptible to ice, and when they are most needed, they may become unusable.

(2) There are two sizes of venturi tubes: those which produce four inches of suction are used to drive the attitude gyros, and smaller tubes, which produce two inches of suction, are used for the turn and slip indicator. Some installations use two of the larger venturi tubes connected in parallel to the two attitude gyros, and the turn and slip indicator is connected to one of these instruments with a needle valve between them. A suction gage is temporarily connected to the turn and slip indicator, and the aircraft is flown so the needle valve can be adjusted to the required suction at the instrument when the aircraft is operated at its cruise speed. (See figure 12-1.)

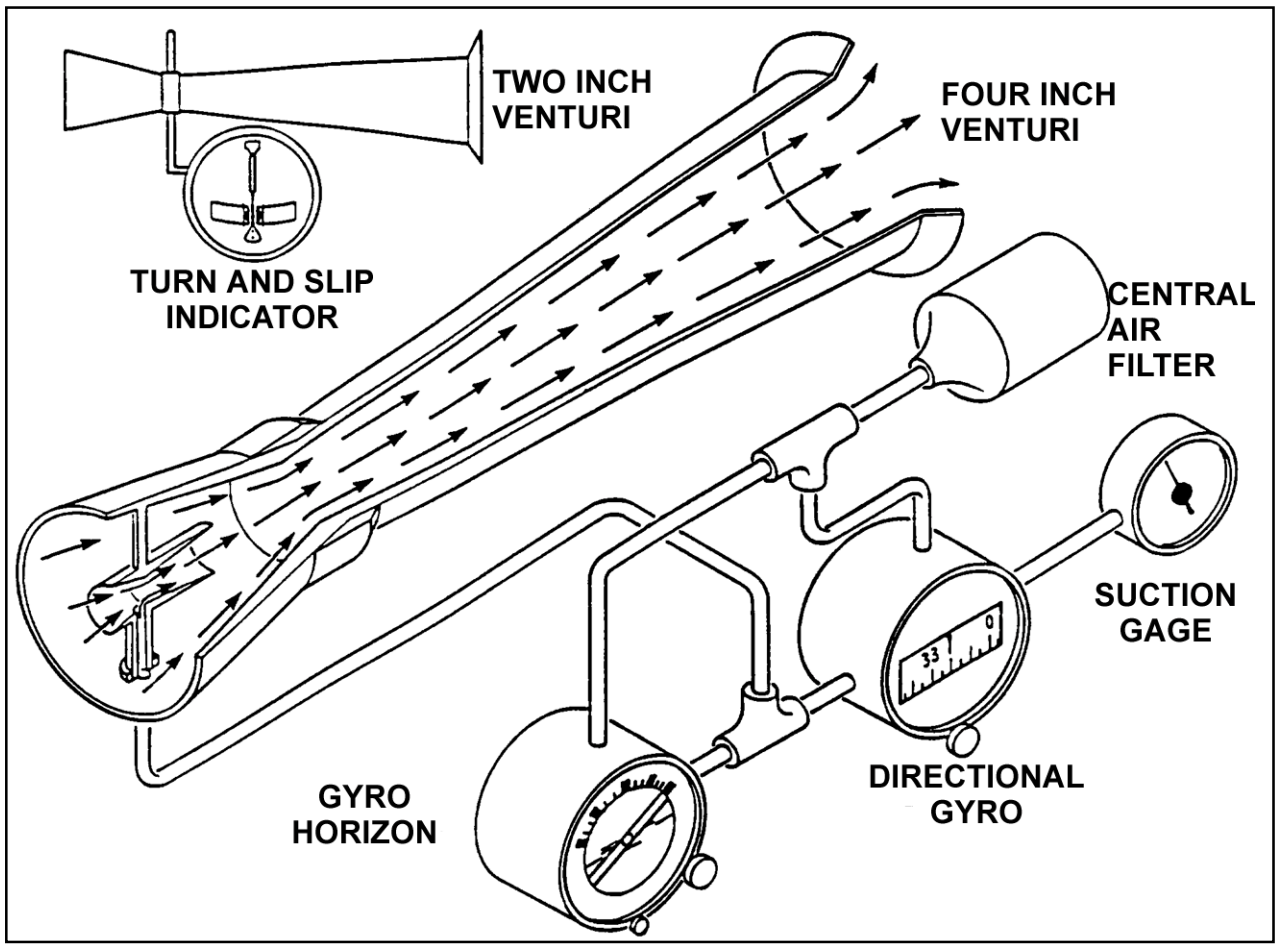


FIGURE 12-1. Venturi system for providing airflow through gyro instruments.

b. Vacuum Pump Systems. In order to overcome the major drawback of the venturi tube, that is, its susceptibility to ice, aircraft were equipped with engine driven vacuum pumps and the gyro instruments were driven by air pulled through the instrument by the suction produced by these pumps. A suction relief valve maintained the desired pressure (usually about four inches of mercury) on the attitude gyro instruments, and a needle valve between one of the attitude indicators and the turn and slip indicator restricted the airflow to maintain the desired 2 inches of suction in its case. Most of the early instruments used only paper filters in each of the instrument cases, but in some installations a central air filter was used to remove contaminants from the cabin air before it entered the instrument case.

(1) The early vacuum pumps were vane-type pumps of what is called the *wet* type-one with a cast iron housing and steel vanes. Engine oil was metered into the pump to provide sealing, lubrication, and cooling, and then this oil, along with the air, was blown through an oil separator where the oil collected on baffles and was returned to the engine crankcase. The air was then exhausted overboard. Aircraft equipped with rubber deicer boots used this discharge air to inflate the boots. But before it could be used, this air was passed through a second stage of oil separation and then to the distributor valve and finally to the boots. (See figure 12-2.)

(2) The airflow through the instruments is controlled by maintaining the suction in the instrument case at the desired level with a suction relief valve mounted between the pump and the instruments. This valve has a spring-loaded poppet that offsets to allow cabin air to enter the pump and maintain the correct negative pressure inside the instrument case.

(3) The more modern vacuum pumps are of the dry type. These pumps use carbon vanes and do not require any lubrication, as the vanes provide their own lubrication as they wear away at a carefully predetermined rate. Other than the fact that they do not require an oil separator, systems using dry air pumps are quite similar to those using a wet pump. One slight difference, however, is in the need for keeping the inside of the pump perfectly clean. Any solid particles drawn into the system through the suction relief valve can damage one of the carbon vanes, and this can lead to destruction of the pump, as the particles broken off of one vane will damage all of the other vanes. To prevent particles entering the relief valve, its air inlet is covered with a filter, and this must be cleaned or replaced at the interval recommended by the aircraft manufacturer.

c. Positive Pressure Systems. Above about 18,000 feet there is not enough mass to the air drawn through the instruments to provide sufficient rotor speed, and, to remedy this problem, many aircraft that fly at high altitude use positive pressure systems to drive the gyros. These systems use the same type of air pump as is used for vacuum systems, but the discharged air from the pump is filtered and directed into the instrument case through the same fitting that receives the filtered air when the vacuum system is used. A filter is installed on the inlet of the pump, and then, before the air is directed into the instrument case, it is again filtered. A pressure regulator is located between the pump and the in-line filter to control the air pressure so only the correct amount is directed into the instrument case.

System Filters. The life of an air-driven gyro instrument is determined to a great extent by the cleanliness of the air that flows over the rotor. In vacuum systems, this air is taken

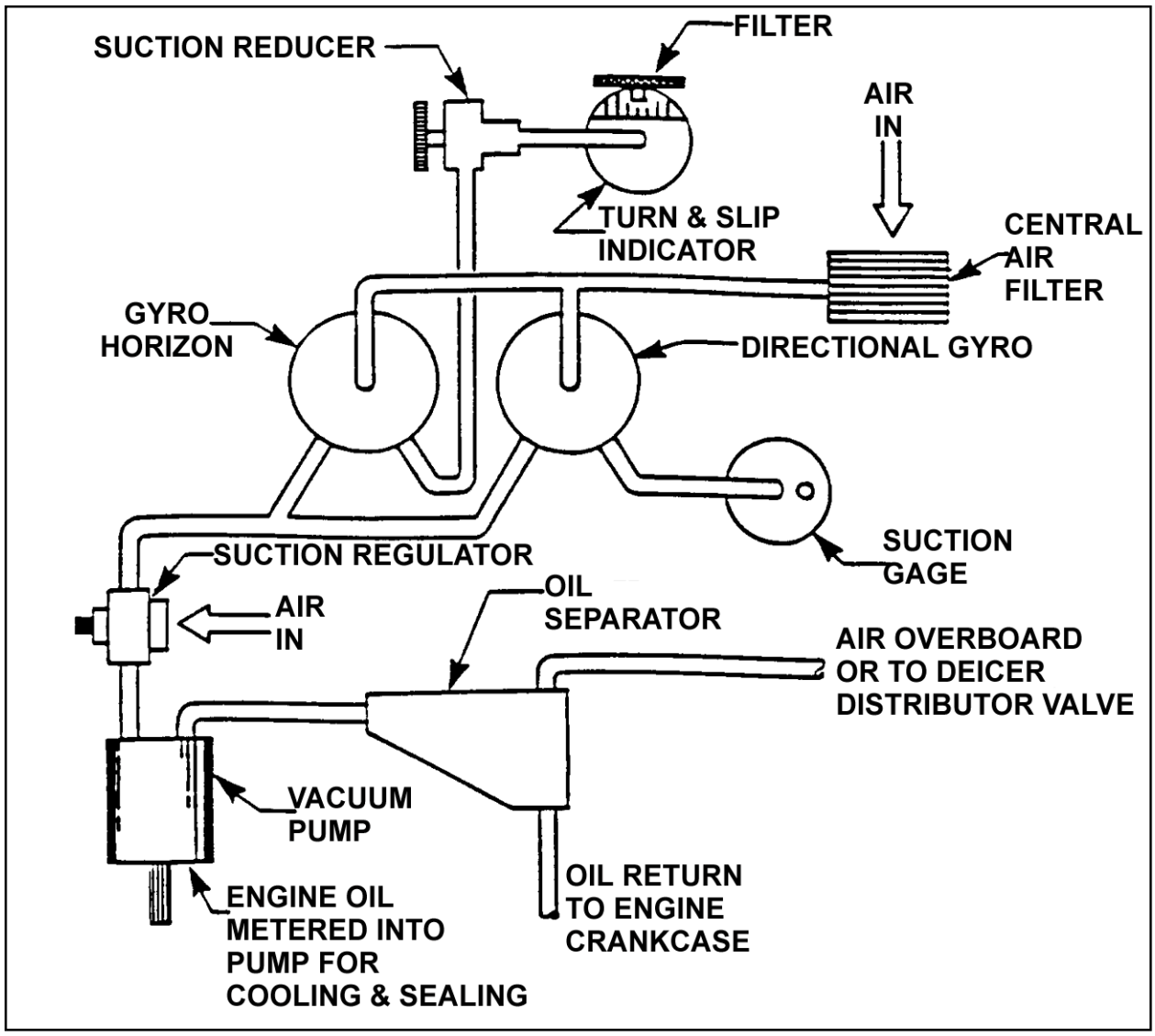


FIGURE 12-2. Instrument vacuum system using a wet-type vacuum pump.

from the cabin where there is usually a good deal of dust and very often tobacco smoke. Unless all of the solid contaminants are removed from the air before it enters the instrument, they will accumulate, usually in the rotor bearings, and slow the rotor. This causes an inaccurate indication of the instrument and will definitely shorten its service life. Dry air pumps are also subject to damage from ingested contaminants, and all of the filters in the system must be replaced on the schedule recommended by the aircraft manufacturer, and more often if the aircraft is operated under particularly dusty conditions, especially if the occupants of the aircraft regularly smoke while flying. (See figures 12-3 and 12-4.)

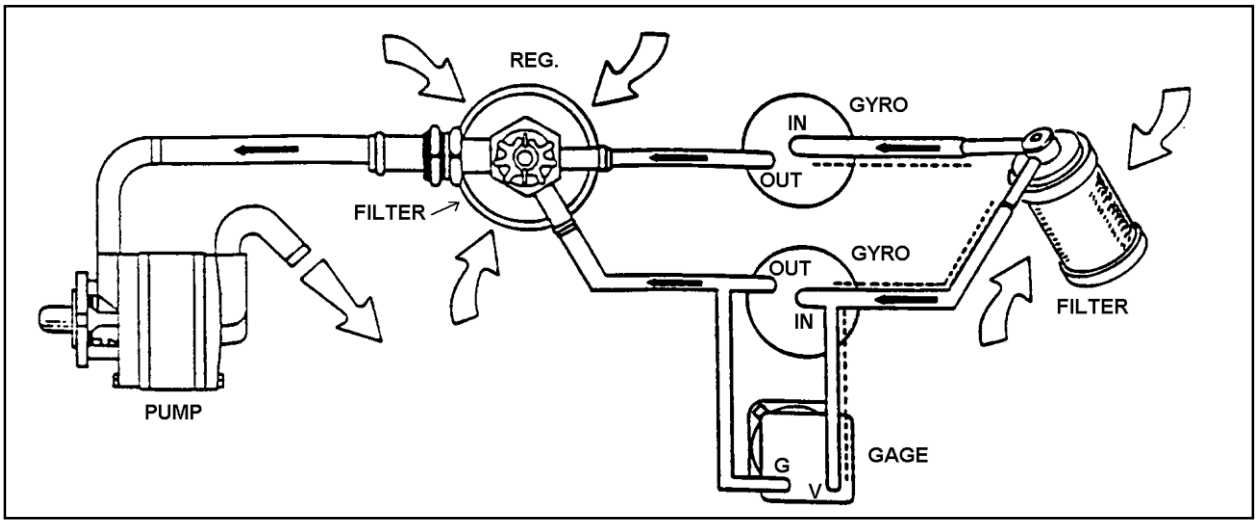


FIGURE 12-3. Instrument vacuum system using a dry-type air pump.

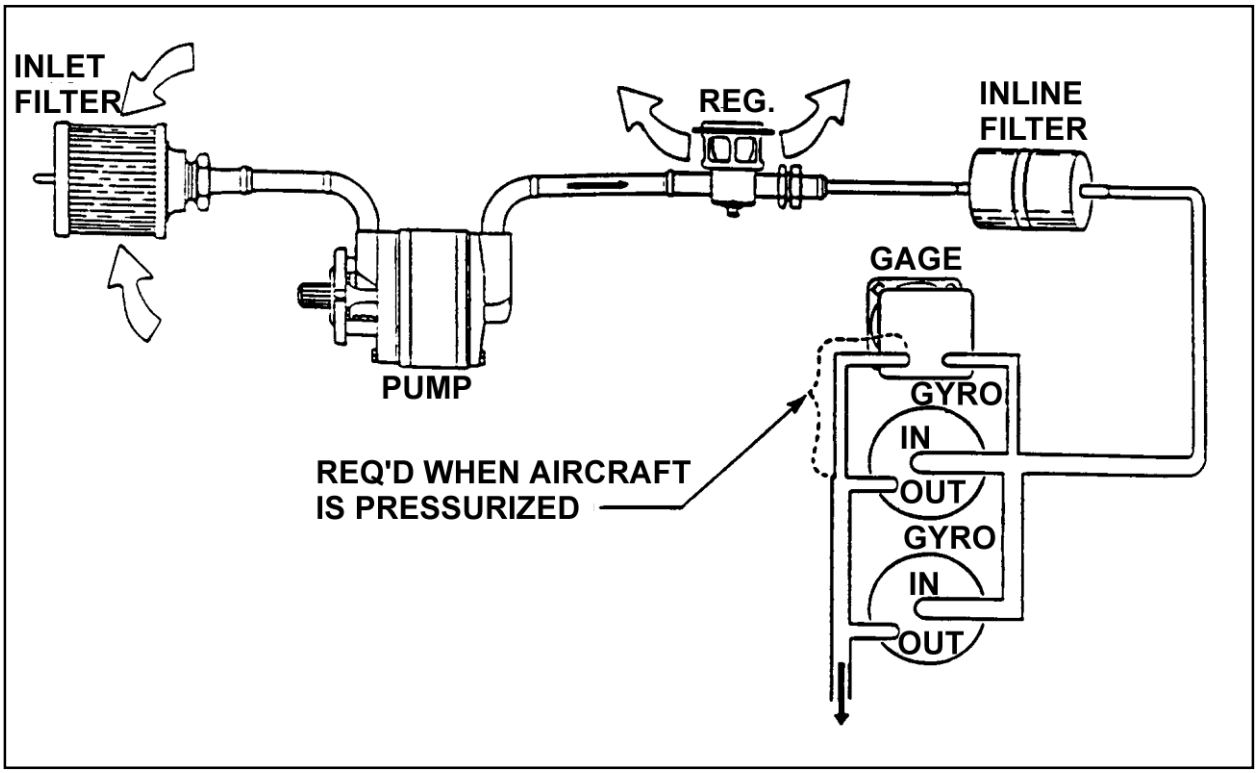


FIGURE 12-4. Instrument pressure system using a dry-type air pump.

12-39.—12-50. [RESERVED.]

SECTION 4. PITOT/STATIC SYSTEMS

12-51. GENERAL. In order for the pitot-static instruments to work properly, they must be connected into a system that senses the impact air pressure with minimum distortion and picks up undisturbed static air pressure.

Pitot pressure is ram air pressure picked up by a small open-ended tube about a ¼-inch in diameter that sticks directly into the air stream that produces a pressure proportional to the speed of the air movement. Static pressure is the pressure of the still air used to measure the altitude and serves as a reference in the measurement of airspeed.

Airspeed requires pitot, altimeter, rate of climb, and transponder-required static.

12-52. SYSTEM COMPONENTS. The conventional design of the pitot system consists of pitot-static tubes or pitot tubes with static pressure parts and vents, lines, tubing, water drains and traps, selector valves, and various pressure-actuated indicators or control units such as the altimeter, airspeed and rate-of-climb indicators, and the encoding altimeter connected to the system. (See figure 12-5.)

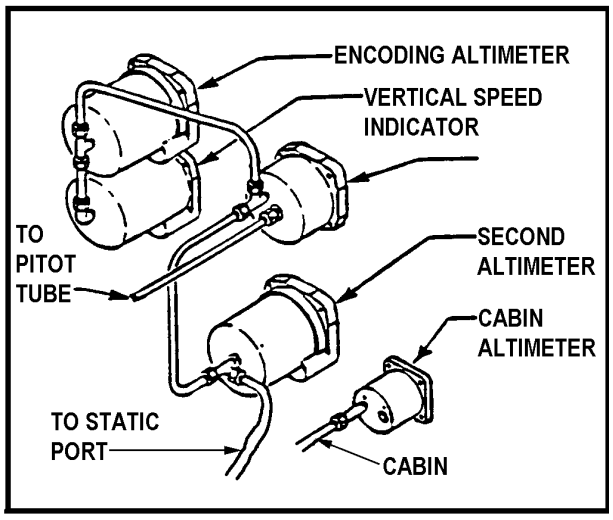


Figure 12-5. Pitot/static system for a small aircraft.

12-53. PITOT/STATIC TUBES AND LINES. The pitot tube (see figure 12-6) is installed at the leading edge of the wing of a single-engine aircraft, outside the propeller slipstream or on the fuselage of a multiengine aircraft with the axis parallel to the longitudinal axis of the aircraft, unless otherwise specified by the manufacturer.

12-54. STATIC PORTS AND VENTS (more modern trend) should be mounted flush with fuselage skin. One port is located on either side of the fuselage, usually behind the cabin.

Inspect for elevation or depression of the port or vent fitting. Such elevation or depression may cause airflow disturbances at high speeds and result in erroneous airspeed and altitude indications.

12-55. HEATER ELEMENTS. A heating element is located within the tube head to prevent the unit from becoming clogged during icing conditions experienced during flight. A switch in the cockpit controls the heater. Some pitot-static tubes have replaceable heater elements while others do not. Check the heater element or the entire tube for proper operation by noting either ammeter current or that the tube or port is hot to the touch. (See figure 12-6.)

12-56. SYSTEM INSPECTION.

a. Inspect air passages in the systems for water, paint, dirt or other foreign matter. If water or obstructive material has entered the system, all drains should be cleaned. Probe the drains in the pitot tube with a fine wire to remove dirt or other obstructions. The bottom static openings act as drains for the head's static chamber. Check these holes at regular intervals to preclude system malfunctioning.

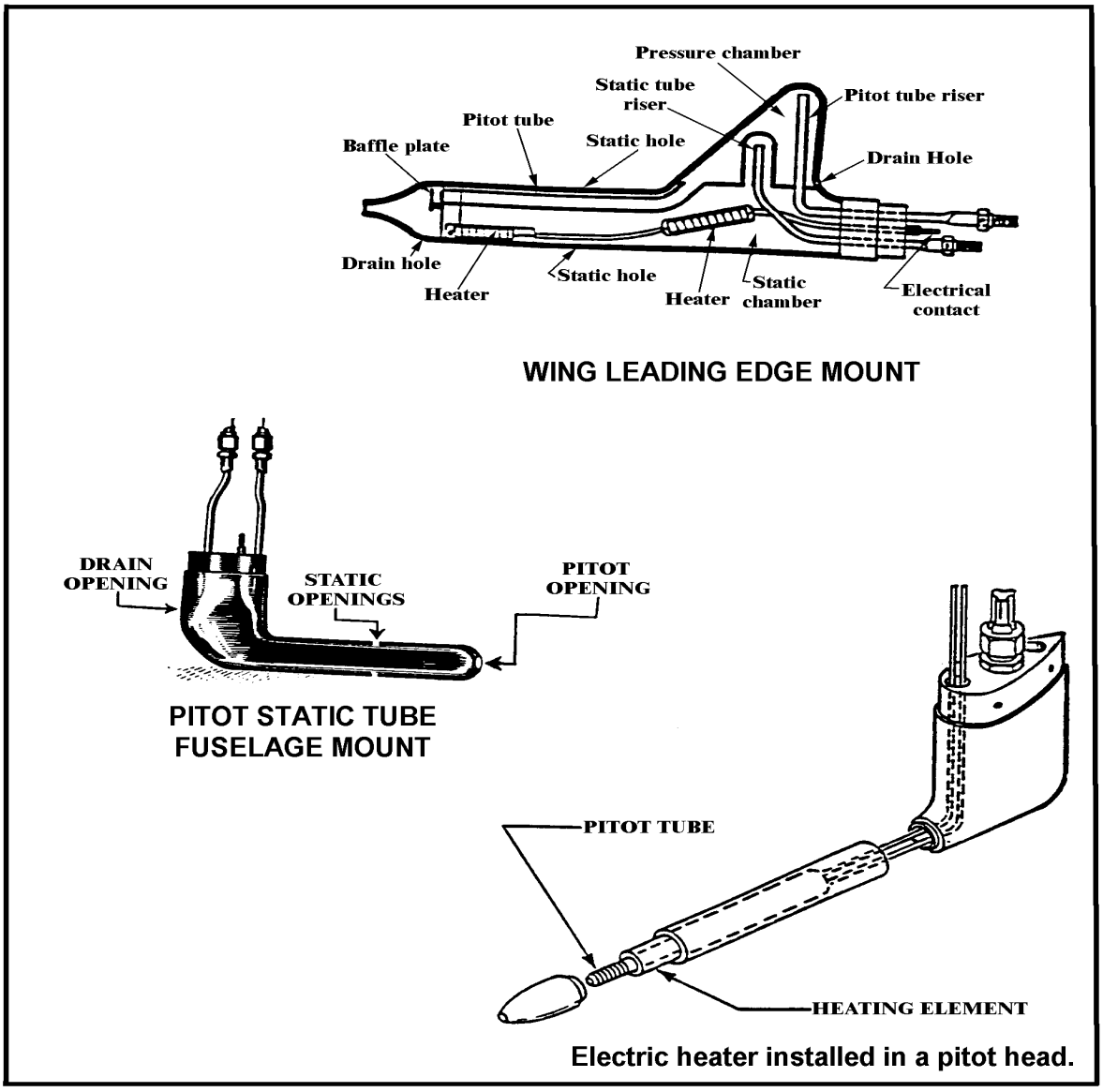


Figure 12-6. Pitot/Static Tube Head

b. Check to ensure the water drains freely. If a problem is experienced with the pitot-static system drainage or freezing at altitude, and the tubing diameter is less than 3/8-inch, replace it with larger tubing.

c. Check the pitot tube for corrosion.

(1) The pitot probe should not have any corrosion within ½-inch of the probe tip.

(2) Make sure there is no flaking which forms pits and irregularities in the surface of the tube.

NOTE: It is essential that the static air system be drained after the airplane has been exposed to rain.

12-57. SYSTEM LEAK TEST.

a. Pitot-static leak tests should be made with all instruments connected to assure that no leaks occur at instrument connections. Such tests should be accomplished whenever a connection has been loosened or an instrument replaced.

b. After the conclusion of the leak test, return the system to its normal flying configuration. Remove tape from static ports and pitot drain holes and replace the drain plug.

12-58. STATIC SYSTEM TESTS must comply with the static system tests required by 14 CFR 91.411 and be performed by an appropriately-rated repair station with the appropriate test equipment.

If the manufacturer has not issued instructions for testing static systems, the following may be used:

a. Connect the test equipment directly to the static ports, if practicable. Otherwise, connect to a static system drain or tee connection and seal off the static ports. If the test equipment is connected to the static system at any point other than the static port, it should be made at a point where the connection may be readily inspected for system integrity. Observe maintenance precautions given in paragraph 12-60 of this section.

b. Do not blow air through the line toward the instrument panel. This may seriously damage the instruments. Be sure to disconnect the instrument lines so no pressure can reach the instruments.

c. Apply a vacuum equivalent to 1,000 feet altitude, (differential pressure of approximately 1.07 inches of mercury or 14.5 inches of water) and hold.

d. After 1 minute, check to see that the leak has not exceeded the equivalent of 100 feet of altitude (decrease in differential pressure of approximately 0.0105 inches of mercury or 1.43 inches of water).

12-59. TEST PITOT SYSTEM in accordance with the aircraft manufacturer's instructions. If the manufacturer has not issued instructions for testing pitot systems, the following may be used:

a. Seal the drain holes and connect the pitot pressure openings to a tee to which a source of pressure and manometer or reliable indicator is connected.

b. Restrain hoses that can whip due to applied pressure.

c. Apply pressure to cause the airspeed indicator to indicate 150 knots (differential pressure 1.1 inches of mercury or 14.9 inches of water), hold at this point and clamp off the source of pressure. After 1 minute, the leakage should not exceed 10 knots (decrease in differential pressure of approximately 0.15 inches of mercury or 2.04 inches of water).

CAUTION: To avoid rupturing the diaphragm of the airspeed indicator, apply pressure slowly and do not build up excessive pressure in the line. Release pressure slowly to avoid damaging the airspeed indicator.

d. If the airspeed indicator reading declines, check the system for leaky hoses and loose connections.

e. Inspect the hoses for signs of deterioration, particularly at bends and at the connection points to the pitot mast and airspeed indicator. Replace hoses that are cracked or hardened with identical specification hoses. Any time a hose is replaced, perform a pressure check.

Warning: Do not apply suction to pitot lines.

12-60. MAINTENANCE PRECAUTIONS. Observe the following precautions in all pitot-static system leak testing:

a. Before any pitot/static system is tested, determine that the design limits of instruments attached to it will not be exceeded during the test. To determine this, locate and identify all instruments attached to the system.

b. A system diagram will help to determine the location of all instruments as well as locate a leak while observing instrument indications. If a diagram is not available, instruments can be located by tracing physical installation.

c. Be certain that no leaks exist in the test equipment.

d. Run full range tests only if you are thoroughly familiar with the aircraft instrument system and test equipment.

e. Make certain the pressure in the pitot system is always equal to, or greater than, that in the static system. A negative differential pressure across an airspeed indicator can damage the instrument.

f. The rate of change or the pressure applied should not exceed the design limits of any pitot or static instruments connected to the systems.

g. When lines are attached to or removed from the bulkhead feed-through fitting or at a union, ensure the line attached to the opposite end is not loose, twisted, or damaged by rotation of the fitting. Such fittings normally are provided with a hex flange for holding the fitting.

12-61. REPLACING LINES. **If necessary to replace lines**, observe the following installation:

a. Attach lines at regular intervals by means of suitable clamps.

b. Do not clamp lines at end fittings.

c. Maintain the slope of lines toward drains to ensure proper drainage.

d. Check the lines for leaks.

12-62. RELOCATON OF PITOT TUBE. If pitot tube relocation is necessary, perform the relocation in accordance with the manufacturer's recommendations and consider the following:

a. Freedom of aerodynamic disturbances caused by the aircraft.

b. Location protected from accidental damage.

c. Alignment with the longitudinal axis of the aircraft when in cruising flight.

12-63. TROUBLESHOOTING THE PITOT/STATIC PRESSURE SYSTEM.

a. If instruments are inoperative or erratic operation occurs, take the following action:

Table 12-1. Color codes for pitot-static systems.

CODE ABBR.	DEFINITION	COLOR
PP	PITOT PRESSURE	NATURAL
SP	STATIC PRESSURE (PILOT)	RED
	STATIC PRESSURE (CO-PILOT)	GREEN
	STATIC PRESSURE (CABIN)	YELLOW
	STATIC PRESSURE (STANDBY)	BLUE

(1) Check for clogged lines. Drain lines at the valves (especially after aircraft has been exposed to rain). Disconnect lines at the instruments and blow them out with low-pressure air.

(2) Check lines for leaks or looseness at all connections. Repair as required.

b. If the pitot heating element(s) are operative, check the following:

(1) Are circuit breaker(s) tripped?

(2) Reset the circuit breaker to determine if:

(a) The system is OK, or

(b) The circuit breaker trips again, if so:

1. Check the wiring continuity to the ground. If the switch(s) is defective, repair as necessary.

2. Check the heating element; replace it if it is defective.

12-64.—12-69. [RESERVED.]

SECTION 5. AVIONICS TEST EQUIPMENT

12-70 GENERAL. Certificated individuals who maintain airborne avionics equipment must have test equipment suitable to perform that maintenance.

12-71 TEST EQUIPMENT CALIBRATION STANDARDS.

a. The test equipment calibration standards must be derived from and traceable to one of the following:

(1) The National Institute of Standards and Technology.

(2) Standards established by the test equipment manufacturer.

(3) If foreign-manufactured test equipment, the standards of the country, where it was manufactured, if approved by the Administrator.

b. The technician must make sure that the test equipment used for such maintenance is the equipment called for by the manufacturer or equivalent.

(1) Before acceptance, a comparison should be made between the specifications of the test equipment recommended by the manufacturer and those proposed by the repair facility.

(2) The test equipment must be capable of performing all normal tests and checking all parameters of the equipment under test. The level of accuracy should be equal to or better than that recommended by the manufacturer.

(3) For a description of avionics test equipment used for troubleshooting, refer to the equipment or aircraft manufacturing instruction manual.

12-72 TEST EQUIPMENT CALIBRATION. Test equipment such as meters, torque wrenches, static, and transponder test equipment should be checked at least once a year.

c. National Institute of Standards and Technology traceability can be verified by reviewing test equipment calibration records for references to National Institute of Standards and Technology test report numbers. These numbers certify traceability of the equipment used in calibration.

d. If the repair station uses a standard for performing calibration, that calibration standard cannot be used to perform maintenance.

e. The calibration intervals for test equipment will vary with the type of equipment, environment, and use. The accepted industry practice for calibration intervals is usually one year. Considerations for acceptance of the intervals include the following:

(1) Manufacturer's recommendation for the type of equipment.

(2) Repair facility's past calibration history, as applicable.

f. If the manufacturer's manual does not describe a test procedure, the repair station must coordinate with the manufacturer to develop the necessary procedures, prior to any use of the equipment.

12-73 – 12-83. [RESERVED.]

CHAPTER 13. HUMAN FACTORS

13-1. HUMAN FACTORS INFLUENCE ON MECHANIC'S PERFORMANCE. To accomplish any task in aviation maintenance at least three things must be in evidence. A mechanic must have the tools, data, and technical skill to perform maintenance. Only recently has the aviation industry addressed the mechanic job functions, pressures, and stress, by identifying those human factors (HF) that impact the mechanics performance.

13-2. THE FAA AVIATION SAFETY PROGRAM has condensed these HF reports into a personal minimums checklist, which asks the mechanic to answer 10 'yes or no' questions before the maintenance task is begun and 10 'yes or no' questions after the task is completed. If the mechanic answers "NO" to any of the 20 questions, the aircraft should not be returned to service. We have provided the checklist in figure 13-1 for your evaluation and review. A color copy of the checklist is available from any Flight Standards District Office. Just ask for the Airworthiness Safety Program manager.

Airworthiness Aviation Safety Program
Federal Aviation Administration

MAINTENANCE

"PERSONAL MINIMUMS" CHECKLIST

√ Before the task

	DO I HAVE THE KNOWLEDGE TO PERFORM THE TASK?
	DO I HAVE THE TECHNICAL DATA TO PERFORM THE TASK?
	HAVE I PERFORMED THE TASK PREVIOUSLY?
	DO I HAVE THE PROPER TOOLS AND EQUIPMENT TO PERFORM THE TASK?
	HAVE I HAD THE PROPER TRAINING TO SUPPORT THE JOB TASK?
	AM I MENTALLY PREPARED TO PERFORM THE JOB TASK?
	AM I PHYSICALLY PREPARED TO PERFORM THE TASK?
	HAVE I TAKEN THE PROPER SAFETY PRECAUTIONS TO PERFORM THE TASK?
	DO I HAVE THE RESOURCES AVAILABLE TO PERFORM THE TASK?
	HAVE I RESEARCHED THE FAR'S TO ENSURE COMPLIANCE?

Airworthiness Aviation Safety Program
Federal Aviation Administration

√After the task

	DID I PERFORM THE JOB TASK TO THE BEST OF MY ABILITIES?
	WAS THE JOB TASK PERFORMED TO BE EQUAL TO THE ORIGINAL?
	WAS THE JOB TASK PERFORMED IN ACCORDANCE WITH APPROPRIATE DATA?
	DID I USE ALL THE METHODS, TECHNIQUES, AND PRACTICES ACCEPTABLE TO INDUSTRY?
	DID I PERFORM THE JOB TASK WITHOUT PRESSURES, STRESS, AND DISTRACTIONS?
	DID I REINSPECT MY WORK OR HAVE SOMEONE INSPECT MY WORK BEFORE RETURNING TO SERVICE?
	DID I MAKE THE PROPER RECORD ENTRIES FOR THE WORK PERFORMED?
	DID I PERFORM THE OPERATIONAL CHECKS AFTER THE WORK WAS COMPLETED?
	AM I WILLING TO SIGN ON THE BOTTOM LINE FOR THE WORK PERFORMED?
	AM I WILLING TO FLY IN THE AIRCRAFT ONCE IT IS APPROVED FOR THE RETURN TO SERVICE?

FIGURE 13-1. Personal Minimum's Checklist

APPENDIX 1. GLOSSARY

The following words and terms represent some of those that are often encountered in the field of aviation. For a more complete list of definitions, a mechanic or technician should consult an aviation dictionary.

abrasion resistant PTFE—a solid insulation wall of PTFE with hard, nonconductive grit positioned midway in the wall thickness, and significantly improves the resistance of the PTFE material to damage from wear.

acetylene—gas composed of two parts of carbon and two parts of hydrogen. When burned in the atmosphere of oxygen, it produces one of the highest flame temperatures obtainable.

acetylene regulator—manually adjustable device used to reduce cylinder pressure to torch pressure and to keep the pressure constant. They are never to be used as oxygen regulators.

adherend—one of the members being bonded together by adhesive.

Airworthiness Directive—a regulation issued by the FAA that applies to aircraft, aircraft engines, propellers, or appliances, when an unsafe condition exists and that condition is likely to exist or develop in other products of the same type design.

airworthy—is when an aircraft or one of its component parts meets its type design and is in a condition for safe operation.

ambient light—the visible light level measured at the surface of the part.

ampere (A)—the basic unit of current flow. One A is the amount of current that flows when a difference of potential of 1 V is applied to a circuit with a resistance of 1 Ω. One coulomb per second.

antenna—a device designed to radiate or intercept electromagnetic waves.

anti-tear strips—strips of fabric of the same material as the airplane is covered with, laid over the wing rib under the reinforcing tape.

apparent power—the product of volts and amperes in AC circuits where the current and voltage are out of phase.

appliance—any instrument, mechanism, equipment, part, apparatus, appurtenance, or accessory, including communications equipment, that is used or intended to be used in operating or controlling an aircraft in flight, is installed in or attached to the aircraft, and is not part of an airframe, engine or propeller.

arm—a measurement of distance, in inches, feet, etc., used in weight and balance calculations. Normally only the longitudinal arm is of practical importance. The three axial arms are longitudinal arm, lateral arm, and vertical arm.

automatic direction finder (ADF)—a radio receiver utilizing a directional loop antenna that enables the receiver to indicate the direction from which a radio signal is being received; also called a radio compass.

automatic flight control system (AFCS)—a flight control system incorporating an automatic pilot with additional systems such as a VOR coupler, an ILS approach coupler, and an internal navigation system that is fully automatic, so the aircraft can be flown in a completely automatic mode.

avionics—the science and technology of electronics as applied to aviation.

azimuth—angular distance measured on a horizontal circle in a clockwise direction from either north or south.

balance—the condition of stability which exists in an aircraft when all weight and forces are acting in such a way as to prevent rotation about an axis or pivot point.

base metal—the metal to be welded, brazed, soldered, or cut.

black light—electromagnetic radiation in the near ultraviolet range of wavelength.

blade station—is a reference position on a blade that is a specified distance from the center of the hub.

bond—the adhesion of one surface to another, with or without the use of an adhesive as a bonding agent.

bonding—a general term applied to the process of electrically connecting two or more conductive objects. In aircraft, the purpose of bonding (except as applied to individual connections in the wiring and grounding systems) is to provide conductive paths for electric currents. This is accomplished by providing suitable low-impedance connections joining conductive aircraft components and the aircraft structure. Another purpose of bonding is to ensure the safe passage of current caused by lightning or static electricity through the aircraft structure.

borescope—a long, tubular optical instrument designed for remote visual inspection of surfaces.

brashness—a condition of wood characterized by low resistance to shock and by an abrupt failure across the grain without splintering.

braze welding—a welding process variation in which a filler metal, having a liquidus above 450 °C (840 °F) and below the solidus of the base metal is used. Unlike brazing, in braze welding the filler metal is not distributed in the joint by capillary action.

brazing—the joining of two pieces of metal by wetting their surface with molten alloy of copper, zinc, or tin.

bus or bus bar—solid copper strips to carry current between primary and secondary circuits; also used as jumpers.

butt joint—a joint between two members aligned approximately in the same plane.

butyrate dope—a finish for aircraft fabric consisting of a film base of cellulose fibers dissolved in solvents with the necessary plasticizers, solvent, and thinners.

cable—(electrical)—assembly of one or more conductors within an enveloping protective sheath so constructed as to permit use of conductors separately or in a group.

calibration—a set of operations, performed in accordance with a definite document procedure, which compares the measurements performed by an instrument or standard, for the purpose of detecting and reporting, or eliminating by adjustment, errors in the instrument tested.

center of gravity—that point about which the aircraft would balance if suspended. For field weight and balance purposes/control, the center of gravity is normally calculated only along its longitudinal axis (nose to tail), disregarding both the lateral and vertical location.

certification—implies that a certificate is in existence which certifies or states a qualification.

check—a lengthwise separation of the wood, the greater part of which occurs across the rings of annual growth.

chemical conversion coating (Specification MIL-C-81706)—is a chemical surface treatment used on aluminum alloys to inhibit corrosion and to provide a proper surface for paint finishing.

chord—an imaginary straight line joining the leading and trailing edges of an airfoil.

circuit—a closed path or mesh of closed paths usually including a source of EMF.

circuit breaker—a protective device for opening a circuit automatically when excessive current is flowing through it.

close-grained wood—wood with narrow and inconspicuous annual rings. The term is sometimes used to designate wood having small and closely-spaced pores, but in this sense the term “fine-textured” is more often used.

coil shot—production of longitudinal magnetization accomplished by passing current through a coil encircling the part being inspected.

compass—a device used to determine direction on the Earth’s surface. A magnetic compass utilizes the Earth’s magnetic field to establish direction.

compression wood—identified by its relatively wide annual rings, usually eccentric, and its relatively large amount of summer wood, usually more than 50 percent of the width of the annual rings in which it occurs. Compression wood shrinks excessively lengthwise as compared with normal wood.

conductor—a wire or other material suitable for conducting electricity.

conduit—a rigid metallic or nonmetallic casing, or a flexible metallic casing covered with a woven braid or synthetic rubber used to encase electrical cables.

contact—electrical connectors in a switch, solenoid or relay that controls the flow of current.

control panel—an upright panel, open or closed, where switches, rheostats, meters, etc., are installed for the control and protection of electrical machinery.

corrosion—the electrochemical deterioration of a metal resulting from chemical reaction to the surrounding environment.

crack—is a partial separation of material caused by vibration, overloading, internal stresses, nicks, defective assemblies, fatigue, or rapid changes in temperature.

creepage—is the conducting of electrical current along a surface between two points at different potentials. The current's ability to pass between two points increases with higher voltage and when deposits of moisture or other conductive materials exist on the surfaces.

cross grain—grain not parallel with the axis of a piece. It may be either diagonal or spiral grain or a combination of the two.

cross coat—a double coat of dope or paint. It is sprayed on in one direction, and then immediately after the solvent flash-off, it is sprayed at right angles to the first coat.

cure—to change the properties of a thermosetting resin irreversibly by vulcanization or chemical reaction. May be accomplished by the addition of curing (cross-linking) agents, with or without a catalyst, and with or without heat or pressure.

curing temperature—temperature to which a resin or an assembly is subjected in order to cure the resin.

cutting torch—a device used in gas cutting of metals.

damping—limiting the duration of vibration by either electrical or mechanical means.

data—information that supports and/or describes the original aircraft design, alteration or repair including the following: (1) drawings, sketches, and or photographs; (2) engineering analysis; (3) engineering orders; and (4) operating limitations.

datum—imaginary vertical plane from which all horizontal measurements are made or indicated when the aircraft is in level flight attitude.

derating—is a technique whereby a part is stressed in actual usage at values well below the manufacturer's rating for the part. By decreasing mechanical, thermal, and electrical stresses, the probability of degradation or catastrophic failure is lessened.

direct current electrode negative—the arrangement of direct current arc welding leads in which the work is the positive pole and the electrode is the negative pole of the welding arc.

direct current electrode positive—the arrangement of direct current arc welding leads in which the work is the negative pole and the electrode is the positive pole of the welding arc.

discontinuity—an interruption in the normal physical structure or configuration of a part, such as a crack, lap, seam, inclusion, or porosity.

distal tip—the tip, lens end, of a borescope.

dope—liquid applied to fabric to tauten it by shrinking, strengthen it, and render it airtight by acting as a filler.

Dope-proofing—protecting a surface from the chemicals and chafing qualities of dope and doped fabrics.

drape—the ability of tape and broad goods to conform to a contoured shape.

drip loop—a bundle installation method used to prevent water or other fluid contaminants from running down the wiring into a connector.

dry rot—a term loosely applied to many types of wood decay but especially to that which, when in an advanced stage, permits the wood to be easily crushed to a dry powder. The term is actually a misnomer for any decay, since all fungi require considerable moisture for growth.

dwell time—the total time that a penetrant, emulsifier (or remover), or developer remains on the surface of the test part.

dye penetrant inspection—an inspection method for surface cracks in which a penetrating dye is allowed to enter any cracks present and is pulled out of the crack by an absorbent developer. A crack appears as a line on the surface of the developer.

edge grain—edge-grain lumber has been sawed parallel with the pith of the log and approximately at right angles to the growth rings; that is, the rings form an angle of 45 degrees or more with the surface of the piece.

electricity—one of the fundamental quantities in nature consisting of elementary particles, electrons and protons, which are manifested as a force of attraction or repulsion, and also in work that can be performed when electrons are caused to move; a material agency which, when in motion, exhibits magnetic, chemical, and thermal effects, and when at rest is accompanied by an interplay of forces between associated localities in which it is present.

electromagnet—temporary magnet which is magnetized by sending current through a coil of wire wound around an iron core.

Electromagnetic/Radio Frequency Interference (EMI/RFI)—frequency spectrum of electromagnetic radiation extending from subsonic frequency to X-rays. This term should not be used in place of the term Radio Frequency Interference (RFI). (See radio frequency interference.) Shielding materials for the entire EMI spectrum are not readily available.

electromotive force (EMF)—difference of electrical potential measured in volts.

electron—a negative charge that revolves around the nucleus of an atom; a unit of a negative electrical charge.

electronics—general term that describes the branch of electrical science and technology that treats the behavior and effects of electron emission and transmission.

electron Volt (eV)—a unit of energy equal to the energy aquired by an electron falling though potential differences of one volt, approximately 1.602X 10-19 joule.

emulsion-type cleaner—a chemical cleaner which mixes with water or petroleum solvent to form an emulsion (a mixture which will separate if allowed to stand). It is used to loosen dirt, soot, or oxide films from the surface of an aircraft.

epoxy—one of various usually thermosetting resins capable of forming tight cross-linked polymer structures marked by toughness, strong adhesion, high corrosion, and chemical resistance, used especially in adhesives and surface coating.

epoxy primer—a two-part catalyzed material used to provide a good bond between a surface and a surface coating.

epoxy resin—a common thermosetting resin which exhibits exceptionally good adhesion, low cure shrinkage, and low water-absorption properties.

erosion—loss of metal from metal surfaces by the action of small particles such as sand or water.

ETFE—(Frequently referred to by the trade name, *TEFZEL*) a copolymer of PTFE and polyethylene.

exciter—small generator for supplying direct current to the alternator's field windings.

exfoliation corrosion—a form of intergranular corrosion that attacks extruded metals along their layer-like grain structure.

expandable sleeving—open-weave braided sleeving used to protect wire and cables from abrasion and other hazards (commonly known by trade name *EXPANDO*).

FEP—fluorinated ethylene propylene (commonly known by the trade name, *TEFLON*). A melt extrudable fluorocarbon resin, very similar in appearance and performance to PTFE, but with a maximum temperature rating of 200 °C.

ferrous metal—iron, or any alloy containing iron.

fiberglass—the most common material used to reinforce structures in home-built and experimental aircraft. Available as mat, roving, fabric, etc. It is incorporated into both thermoset and thermoplastic resins. The glass fibers increase mechanical strength, impact resistance, stiffness, and dimensional stability of the matrix.

fill—threads in a fabric that run crosswise of the woven material.

filiform corrosion—a thread, or filament-like corrosion which forms on aluminum skins beneath the finish.

finish—external coating or covering of an aircraft or part.

flat grain—lumber has been sawed parallel with the pith of the log and approximately tangent to the growth rings; that is, the rings form an angle of less than 45 degrees with the surface of the piece.

fluorescent—a substance is said to be fluorescent when it will glow or fluoresce when excited by ultraviolet light. Some types of dye-penetrant material use fluorescent dyes which are pulled from the cracks by a developer and observed under "black" ultraviolet light.

flux—materials used to prevent, dissolve, or facilitate removal of oxides and other undesirable surface substances. Also, the name for magnetic fields.

fretting corrosion—corrosion damage between close-fitting parts which are allowed to rub together. The rubbing prevents the formation of protective oxide films and allows the metals to corrode.

fuse—a protective device containing a special wire that melts when current exceeds the rated value for a definite period.

functional check—this test may require the use of appropriate test equipment.

galvanic corrosion—corrosion due to the presence of dissimilar metals in contact with each other.

gas cylinder—a portable container used for transportation and storage of a compressed gas.

gas tungsten arc welding—(GTAW) an arc welding process which produces coalescence of metals by heating them with an arc between a tungsten (nonconsumable) electrode and the work. Shielding is obtained from a gas or gas mixture. Pressure may or may not be used and filler metal may or may not be used.

generator—a device for converting mechanical energy into electrical energy.

global positioning system (GPS)—a navigation system that employs satellite transmitted signals to determine the aircraft's location.

grain—the direction, size, arrangement, appearance, or quality of the fibers in wood or metal.

grain - diagonal—annual rings in wood at an angle with the axis of a piece as a result of sawing at an angle with the bark of the tree.

grommet—an insulating washer that protects the sides of holes through which wires must pass/or a metal or plastic drain attached to fabric on aircraft.

gross weight —the total weight of the aircraft including its contents.

grounding—the term is usually applied to a particular form of bonding that is the process of electrically connecting conductive objects to either conductive structure or some other conductive return path for the purpose of safely completing either a normal or fault circuit.

harness—a cable harness is a group of cables or wires securely tied as a unit.

honeycomb—manufactured product consisting of a resin-impregnated sheet or metal material which has been corrugated or expanded into hexagon-shaped and other structural-shaped cells. Primarily used as core material for sandwich constructions.

inductance (L)—the ability of a coil or conductor to oppose a change in current flow.

insulator—a material that will not conduct current to an appreciable degree.

integrated circuit—small, complete circuit built up by vacuum deposition and other techniques, usually on a silicon chip, and mounted in a suitable package.

intergranular corrosion—the formation of corrosion along the grain boundaries within a metal alloy.

interlocked-grained wood—wood in which the fibers are inclined in one direction in a number of rings of annual growth, then gradually reverse and are inclined in an opposite direction in succeeding growth rings, then reverse again.

inverter—a device for converting direct current to alternating current.

laminate—a product obtained by bonding two or more laminae of the same material or of different materials.

laminated wood—a piece of wood built up of plies or laminations that have been joined either with glue or with mechanical fastenings. The term is most frequently applied where the plies are too thick to be classified as veneer and when the grain of all plies is parallel.

leakage field—the magnetic field forced out into the air by the distortion of the field within a part, caused by the presence of a discontinuity or change in section configuration.

linter—the short fiber left on the cotton seed after ginning.

localizer—that section of an ILS that produces the directional reference beam.

LORAN (Long-Range Navigation)—a radio navigation system utilizing master and slave stations transmitting timed pulses. The time difference in reception of pulses from several stations establishes a hyperbolic line of position that may be identified on a LORAN chart. By utilizing signals from two pairs of stations, a fix in position is obtained.

magnetic field—the space around a source of magnetic flux in which the effects of magnetism can be determined.

marker beacon—a radio navigation aid used in an instrument approach to identify distance to the runway. As the aircraft crosses over the marker-beacon transmitter, the pilot receives an accurate indication of the airplane's distance from the runway through the medium of a flashing light and an aural signal.

master switch—a switch designed to control all electric power to all circuits in a system.

moisture content of wood—weight of the water contained in the wood usually expressed in percentage of the weight of the kiln-dry wood.

multiconductor cable—consists of two or more cables or wires, all of which are encased in an outer covering composed of synthetic rubber, fabric, or other material.

nick—a sharp notch-like displacement of metal surface.

nomex braid—*NOMEX* is the trade name for a high-temperature polyamide thread that is braided over the larger sizes (# 8 gage and larger) of many of the military specification wires. It can be encountered in either an off-white or black/green color.

normalizing—reforming of the grain structure of a metal or alloy by proper heat treatment to relieve internal stresses.

open circuit—an incomplete or broken electrical circuit.

open-grained wood—common classification of painters for woods with large pores, such as oak, ash, chestnut, and walnut. Also known as "coarse-textured."

operational check—this is an operational test to determine whether a system or component is functioning properly in all aspects in conformance with minimum acceptable manufacture design specifications.

optical fiber—any filament or fiber made of dielectric materials that guides light whether or not it is used to transmit signals.

rifice—opening through which gas or air flows. It is usually the final opening controlled by a valve.

oxidizing—combining oxygen with any other substance. For example, a metal is oxidized when the metal is burned, i.e., oxygen is combined with all the metal or parts of it

oxidizing flame—an oxy-fuel gas flame having an oxidizing effect due to excess oxygen.

oxygen cutting—cutting metal using the oxygen jet which is added to an oxygen-acetylene flame.

oxygen regulator—manually-adjustable device used to reduce cylinder pressure to torch pressure and to keep the pressure constant. They are never to be used as fuel gas regulators.

peel ply—a layer of resin-free material used to protect a laminate for later secondary bonding (sometimes referred to as a release film).

pickling—the treatment of a metal surface by an acid to remove surface corrosion.

pitch—is the distance, in inches, that a propeller section will move forward in one revolution, or the distance a nut will advance in one revolution of the screw in a single thread.

pitch distribution—is the gradual twist in the propeller blade from shank to tip.

pitted—small irregular shaped cavities in the surface of the parent material usually caused by corrosion, chipping, or heavy electrical discharge.

pitting—the formation of pockets of corrosion products on the surface of a metal.

plastic—an organic substance of large molecular weight which is solid in its finished state and, at some stage during its manufacture or its processing into a finished article, can be shaped by flow.

polyester braid—a plastic braiding thread, when used as the outer surface of a wire, provides a cloth-like appearance.

polyimide tape—a plastic film (commonly referred to by the trade name, *KAPTON*). The tape has a dark brown color, and is frequently coated with a polyimide varnish that has a very distinct mustard yellow color. At times, the spiral edge of the outermost tape is apparent under the varnish topcoat. It may be used for wire insulation. Total polyimide tape insulated wire constructions are inactive for new design on military aircraft and are subject to the procedures defined in FAA Advisory Circular AC 29-2A Change 2 Paragraph 29.1359 in Civil Aircraft.

polyimide varnish—a liquid form of polyimide that is applied to the outer surface of a wire through the process of repeated dipping through the varnish bath with subsequent heat curing. The successive layers rarely reach a total buildup of 1 mil.

polymerization—basic processes for making large (high-polymer) molecules from small ones, normally without chemical change; can be by addition, condensation, rearrangement, or other methods.

porosity—cavity-type discontinuities in metal formed by gas entrapment during solidification.

prepreg—a mat, a fabric, or covering impregnated with resin that is ready for lay up and curing.

propeller—is a rotating airfoil that consists of two or more blades attached to a central hub which is mounted on the engine crankshaft.

protractor—is a device for measuring angles.

PTFE Tape (Insulation)—polytetrafluoroethylene tape (commonly known by the trade name, *TEFLON*), wrapped around a conductor and then centered with heat, fusing the layers into a virtually homogeneous mass. It is used both as a primary insulation against the conductor, and as an outer layer or jacket over a shield. Maximum temperature rating is 260 °C.

PVF_2 Polyvinylidine Fluoride—a fluorocarbon plastic, that when used in aircraft wire, is invariably radiation cross-linked and employed as the outer layer.

radar (radio detecting and ranging)—radio equipment that utilizes reflected pulse signals to locate and determine the distance to any reflecting object within its range.

radome—a nonmetallic cover used to protect the antenna assembly of a radar system.

rectifier—a device for converting alternating current to direct current.

reinforcing tape—a narrow woven cotton or polyester tape used over aircraft fabric to reinforce it at the stitching attachments.

relay—an electrically-operated remote-control switch.

resin—vast profusion of natural and increasingly, synthetic materials used as adhesives, fillers, binders and for insulation.

resistance—the opposition a device or material offers to the flow or current.

resonance method (ringing) of ultrasonic inspection—a method of detecting material thickness or indications of internal damage by injecting variable frequency ultrasonic energy into a material. A specific frequency of energy will produce the clearest indication of damage in a given thickness of material. When the equipment is calibrated for a specific thickness, and this thickness changes, an aural or visual alert is given.

resonant frequency—the frequency of a source of vibration that is exactly the same as the natural vibration frequency of the structure.

resonate—a mechanical system is said to resonate when its natural vibration frequency is exactly the same as the frequency of the force applied. When an object resonates at a particular frequency, the amplitude in its vibration will increase immensely as that frequency is reached and will be less on either side of that frequency.

rib—part of primary structure, whose purpose is to maintain profile of airfoil and support fabric or thin wood covering.

sacrificial corrosion—a method of corrosion protection in which a surface is plated with a metal less noble than itself. Any corrosion will attack the plating rather than the base metal.

sandwich construction—a structural panel concept consisting in its simplest form of two relatively thin, parallel sheets (face sheets) of structural material bonded to and separated by a relatively thick, lightweight core. High strength-to-weight ratios are obtained with sandwiched materials.

scarf joint—a joint made by cutting away similar angular segments of two adherents and bonding the adherents with cut areas fitted together.

score—a surface tear or break on a surface that has a depth and length ranging between a scratch and a gouge.

scratch—a superficial small cut on a surface.

semiconductor device—any device based on either preferred conduction through a solid in one direction, as in rectifiers; or on a variation in conduction characteristics through a partially conductive material, as in a transistor.

severe wind and moisture problem (SWAMP) areas—areas such as wheel wells, wing folds, and near wing flaps, and areas directly exposed to extended weather conditions are considered SWAMP areas on aircraft.

silicone rubber—a high temperature (200 °C) plastic insulation that has a substantial silicone content.

soldering—a group of welding processes that produces coalescence of materials by heating them to the soldering temperature and by using a filler metal having a liquidus not exceeding 450 °C (840 °F) and below the solidus of the base metals. The filler metal is distributed between the closely-fitted surfaces of the joint by capillary action.

solenoid—a tubular coil for the production of a magnetic field; electromagnet with a core which is able to move in and out.

spar—main spanwise structural member(s) of an aircraft wing or rotorcraft rotor. A wing may have one or two made into a single strong box to which secondary leading and trailing structures are added.

spiral grain—a type of growth in wood which the fibers take a spiral course about the bole of a tree instead of the normal vertical course. The spiral may extend right-handed or left-handed around the tree trunk.

stator—the part of an AC generator or motor which contains the stationary winding.

stress corrosion—corrosion of the intergranular type that forms within metals subject to tensile stresses which tend to separate the grain boundaries.

surface tape—pinked-edge strips of fabric doped over all seams, rib stitching, and edges of fabric covering (also called finishing tape).

switch—a device for opening or closing an electrical circuit.

tape—a tape or a “narrow fabric” is loosely defined as a material that ranges in width from 1/4 inch to 12 inches.

TCAS—traffic alert and collision avoidance system. An airborne system that interrogates mode A, C, and S transponders in nearby aircraft and uses the replies to identify and display potential and predicted collision threats.

thermocouple—device to convert heat energy into electrical energy.

thermoplastic material—a material that can be repeatedly softened by an increase in the temperature and hardened by a decrease in the temperature with no accompanying chemical change. For example, a puddle of tar on the road in the summer during the heat of day: the tar is soft and fluid; however, when cooler in the evening, it becomes solid again.

thermoset material—a material which becomes substantially infusible and insoluble when cured by the application of heat or by chemical means. A material that will undergo, or has undergone, a chemical reaction (different from a thermoplastics physical reaction) by the action of heat, catalysts, ultraviolet light, etc. Once the plastic becomes hard, additional heat will not change it back into a liquid as would be the case with a thermoplastic.

tip—part of the torch at the end where the gas burns, producing the high-temperature flame.

transceiver—a unit serving as both a receiver and a transmitter.

transformer—a device for raising or lowering AC voltage.

transmitter—an electronic system designed to produce modulated RF carrier waves to be radiated by an antenna; also, an electric device used to collect quantitative information at one point and send it to a remote indicator electrically.
transponder—an airborne receiver-transmitter designed to aid air traffic control personnel in tracking aircraft during flight.

unbonding—adhesive or cohesive failure between laminates. Compare definitions of adhesive, cohesive debond, and disbond.

very high frequency (VHF)—a frequency between 30 and 300 MHz

VHF omnirange (VOR)—an electronic air navigation system that provides accurate direction information in relation to a certain ground station.

videoscope—a type of borescope.

visible light—electromagnetic radiation that has a wavelength in the range from about 3,900 to 7,700 angstroms and that may be seen by the unaided human eye.

visual check—utilizing acceptable methods, techniques, and practices to determine physical condition and safety item.

volt—unit of potential, potential difference, or electrical pressure.

voltage regulator—device used in connection with generators to keep the voltage constant as load or speed is changed.

warp—threads in a fabric that run the length of the woven material as it comes from the mill.

watt—the unit of power; equal to a joule per second.

wattmeter—an instrument for measuring electrical power.

waveguide—a hollow, typically rectangular, metallic tube designed to carry electromagnetic energy at extremely high frequencies.

wavy-grained wood—wood in which the fibers collectively take the form of waves or undulations.

welding—a materials-joining process used in making welds.

welding rod—a form of welding filler metal, normally packaged in straight lengths.
welding torch—the device used in gas welding.

wood decay—disintegration of wood substance through the action of wood-destroying fungi.

wood decay - incipient—the early stage of decay in which the disintegration has not proceeded far enough to soften or otherwise perceptibly impair the hardness of the wood.

wood decay - typical or advanced—the stage of decay in which the disintegration is readily recognized because the wood has become punky, soft and spongy, stringy, pitted, or crumbly.

x-ray—a radiographic test method used to detect internal defects in a weld.

XL-ETFE—A process of radiation cross-linking the polymer chains is used to thermally set the plastic. This prevents the material from softening and melting at elevated temperature.

XL-Polyalkene—an insulation material based on the polyolefin family that has its normally thermomelt characteristic altered by the radiation cross-linking process to that of a nonmelt, therm-set material.

APPENDIX 2. ACRONYMS AND ABBREVIATIONS

The acronyms and abbreviations listed are some of many that are likely to be encountered by the aviation mechanic or technician involved in the maintenance of aircraft.

429—ARINC 429 data bus standard
629—ARINC 629 data bus standard
A/D—analog/digital; analog-to -digital
A/D CONV—analog-to -digital converter
A/L—autoland
AC—Advisory Circular
ac—alternating current
ACARS—ARINC Communication Addressing and Reporting System
ACO—Aircraft Certification Office
AD—Airworthiness Directive
ADC—air-data computer
ADCP—ATC dual-control panel
ADEDS—advanced electronic display system
ADF—automatic direction finder
ADI—attitude-director indicator; air data instrument
AFC—automatic frequency control
AFCS—automatic flight control system
AFDS—autopilot flight director system
AIM—Aeronautical Information Manual
AIRCOM—air/ground communications
AM—amplitude modulation
AMP or **AMPL**—amplifier
AMP—amperes
AMS—Aerospace Material Specification
AN—Army/Navy
AND—Army Navy Design
ANSI—American National Standards Institute
ANT—antenna
AP—autopilot
APB—auxiliary power breaker
APCU—auxiliary power control unit
APU—auxiliary power unit
ARINC—Aeronautical Radio Incorporated
ARNC IO—ARINC I/O error
ARNC STP—ARINC I/O UART data strip error
ASTM—American Society for Testing Materials
ATA—Air Transport Association
ATC—air traffic control
ATCT—ATC transponder
ATCTS—ATC transponder system
AUX—auxiliary
AVC—automatic volume control
AWG—American Wire Gauge
AWS—Air Weather Service
B/CU—battery/charger unit
BAT or **BATT**—battery
BCD—binary-coded decimal
BIT—binary digit; built-in test
BITE—built-in test equipment
BITS—bus interconnect transfer switch
BNR—binary numerical reference; binary
BP—band-pass
BPCU—bus power control unit
BT—bus tie
BTB—bus tie breaker
BTC—before top center
BUS—electrical bus; 429 digital data bus
C.G.—Center of Gravity
CAC—caution advisory computer
CAGE—commercial and government entity code
CAWS—central aural warning system; caution and warning system
CB, C/B, or **CKT/BKR—**circuit breaker
CDI—course-deviation indicator
CDU—central display unit
CFC—carbon fiber composite
CFDIU—centralized fault display interface unit
CFDS—centralized fault display system
CH or **CHAN—**channel
CHGR—charger
CKT—circuit
CLK—clock
CLR—clear
CMCS—central maintenance computer system
CMPTR—computer
CO—carbon monoxide
COAX—coaxial
COP—copper
CP—control panel
CRT—cathode-ray tube; circuit
CSE or **CSEU —**control system electronics unit
CSEUP—control system electronics unit panel
CT—computed tomography
CT—current transformer
CTN—caution
CU—control unit; copper
CVR—cockpit voice recorder
CW—continuous wave
D/A—digital-to-analog
DAC—digital-to analog converter
DADC—digital air-data computer
DBT—dead bus tie
dc—direct current

DCDR—decoder
DDB—digital data bus
DEMOD—demodulator
DEMUX—demultiplexer
DFDR—digital flight data recorder
DG—directional gyro
DGTL—digital
DH—decision height
DISC SOL—disconnect solenoid
DISC—disconnect
DISTR—distribution
DMA—direct memory access
DMB—dead main bus
DMC—display management computer
DME—distance-measuring equipment
DMEA—distance-measuring equipment antenna
DN—down
DU—display unit
E/E—or E & E electrical/electronic
E1-1—first shelf, number 1 equipment rack
E2-2—second shelf, number 2 equipment rack
EADF—electronic automatic direction finder
EADI—electronic attitude-director indicator
EAROM—electrically alterable read-only memory
EC—EICAS computer
ECAM—electronic centralized aircraft monitoring
EDSP—EICAS display select panel
EDU—EICAS display unit
EEC—electronic engine control
EFI—electronic flight instrument
EFIS—electronic flight instrument system
EFISCP—EFIS control panel
EFISCU—EFIS comparator unit
EFISG EFIS—symbol generator
EFISRLS EFIS—remote light sensor
EHSI—electronic horizontal-situation indicator
EHSID—electronic horizontal-situation indicator display
EHSV—electrohydraulic servo value
EICAS—engine indicating and crew alerting system
ELCU—electrical load control unit
ELEC—electric; electronic
ELECT—electrical
ELEX—electronics; electrical
ELT—Emergency Locator Transmitter
EMER GEN—emergency generator
emf—electromotive force
EMFI—electromechanical flight instrument
EMI—Electromagnetic interference
EP AVAIL—external power available
EP—external power
EPC—external power contactor
EPCS—electronic power control switch
EPROM—erasable programmable read-only memory
eV—electron volt
EXCTR—exciter
EXT PWR—external power
FAA—Federal Aviation Administration
FAA-PMA—Federal Aviation Administration Parts Manufacturer Approval
FM—frequency modulation
FM/CW—frequency modulation continuous wave
FMC—flight management computer
FMCD—flight management computer control display unit
FMCS—flight management computer system
FMS—flight management system
FOD—foreign object damage
FREQ—frequency
FSEU—flap/slat electronic unit
FW or FWD—forward
G/S—glide slope
GAL or GALY—galley
GCR—generator control relay auxiliary contact
GCU—generator control unit
GEB—generator circuit breaker
GEN—generator
GLR—galley load relay
GMAW—gas metal arc welding
GMT—Greenwich mean time; cordinated Universal time
GND PWR—ground power
GND RET—ground return
GND SVCE—ground service
GND or **GRD**—ground
GPCU—ground power control unit
GPS—global positioning system
GPSW—gear opposition switch
GPU—ground power unit
GPW—ground proximity warning
GPWS—ground proximity warning system
GSR—ground service relay
GSSR—ground service select relay
GSTR—ground service transfer relay
GTAW—gas tungsten arc welding
GWPC—ground proximity warning computer
H/L—high/low
HEA—high-frequency radio antenna
HF (hf) —high frequency (3 to 30 MHz)
HFCP—high-frequency radio control panel

HI Z—high impedance
HZ—hertz
I.D.—inner diameter
I/O—input/output
IAPS—integrated avionics processor system
IAS—indicated airspeed
IDG—integrated drive generator
IF—intermediate frequency
IFR—instrument flight rules
IGN—ignition
IIS—integrated instrument system
ILS—instrument landing system
INDL—indicator light
INST—instrument
INSTR—instrument
INTCON—interconnect
INTEC—interface
INTER—interrogation
INTPH—interphone
INV—inverter
IR ILS—receiver
kHz—kilohertz
KSI—thousands of pounds per square inch
kV—kilovolts
kVA—kilovoltamperes
kVAR—kilovoltampere reactive
L-Band—radio frequency band (390 to 1550 MHz)
LCD—liquid-crystal display
LD—load
LED—light-emitting diode
LF (lf) —low frequency (30 to 300 kHz)
LO Z—low impedance
LOC—localizer
LRU—line replaceable unit
LS—loudspeaker
LSB—lower sideband
LSPTM—limit switch position transmitter module
LT—light
LTS— lights
MAC—mean aerodynamic chord
MAN/ELEC—manual/electric
MBA—marker-beacon antenna
MCDP—maintenance control and display panel
MCDU—multipurpose control and display unit
MDE—modern digital electronics
MEC—main equipment center; main engine control
MEG or **MEGA—**million
MEK—methylethylketone
MEM—memory
METO—Maximum except-take off
MF—(mf) medium frequency (300 kHz to 3 MHz)
MHz—megahertz
MIC—microphone
MICRO-P—microprocessor
MIG—metal inert gas
MILLI—one one-thousandth (0.001)
MKR BCN—marker beacon
MS—military standard
MSDS—Material Safety Data Sheets
MSEC—(ms) milliseconds
MSG—message
MTBF—mean time-between-failure
MUX—multiplexer
mV—millivolts
NAS—National Aerospace Standard
NAV—navigation
NC—normally closed; not connected; no connection
NDB—nondirectional beacon
NDI—Nondestructive Inspection
NEG—negative
NSEC— (ns) nanoseconds
NTSB—National Transportation Safety Board
NVM—nonvolatile memory
OAM—original aircraft manufacturer
OBS—omni bearing selection
OC—overcurrent
OEM—original equipment manufacturer
OF—over-frequency
OVV or **OV—**overvoltage
OVVCO or **OVCO**—overvoltage cutout
P-S—parallel to series
PA—passenger address; power amplifier
PARA/SER—parallel to serial
PCU—passenger control unit; power control unit
PFD—permanent-magnet generator
PMA—Parts Manufacturer Approval
POS—positive
POT—potentiometer; plan of test
PR—power relay
PRL—parallel
PROM—programmable read-only memory
PROX—proximity
PSEU—proximity switch electronic unit
PSI—pounds per square inch
PWR—power
PWR SPLY—power supply
QPL—Qualified Products List
QTY—quantity
r-t—receiver-transmitter
RA—radio altimeter; radio altitude
RAD—radio

RAIND radio altimeter indicator
RAM—random-access memory
RART—radio altimeter receiver-transmitter
RAT—ram air turbine
RCCB—remote-control circuit breaker
RCL—recall
RCVR—receiver
RCVR/XMTR—receiver/transmitter
RDMI—radio distance magnetic indicator
RF (rf) —radio frequency
RFI—radio-frequency interference
RLS—remote light sensor
RMI—radio magnetic indicator
rpm—revolution per minute
RTV—room temperature vulcanizing
SAE—Society of Automotive Engineers
SAT—static air temperature
SATCOM—satellite communication
SCR—silicon-controlled rectifier
SDI—source destination identifier
SELCAL—selective calling system
SER DL—serial data link
SG—symbol generator
SITA—Société International de Telecommunications Aeronautiques
SMAW—shielded metal arc welding
SMD—surface mounted device
SNR—signal-to-noise ratio
SOL—solenoid
SOLV—solenoid valve
SOM—start of message
SOT—start of transmission
SPKR—speaker
SPR—software problem report
SQL—squelch
SSB—single sideband
SSID—Supplemental Structural Inspection Documents
SSM—sign status matrix
ST—synchro transmitter
STAT INV—static inverter
STBY—standby
STC—Supplemental Type Certificate
SW—switch
SYM GEN—symbol generator
T-R—transformer-rectifier
TAT—true air temperature
TBDP—tie bus differential protection
TC—Type Certificate
TCAS—traffic alert and collision avoidance system
TCDS—Type Certificate Data Sheets
TDC—top dead center
TFR—transfer
TIG—tungsten inert gas
TMC—thrust management computer
TMS—terminal marking sleeve
TMS—thrust management system
TMSP—thrust mode select panel
TRU—transformer-rectifier unit
TSO—Technical Standard Order
TXPDR—transponder
µ—micro
UBR—utility bus relay
UF—underfrequency
UHF—ultrahigh frequency (300 MHz to 3 GHz)
UNDF—underfrequency
UNDV—undervoltage
US—underspeed
USB (us)—upper sideband
USEC—microseconds
UV—undervoltage
UV—utraviolet
V ac, Vac, or **VAC—**volts alternating current
V dc, Vdc, or **VDC** —volts direct current
V—volts; voltage; vertical; valve
VA—volt-amperes
VAR—volt-ampere reactive
VFR—visual flight rules
VHF (vhf) —very high frequency (30 TO 300 MHz)
VLSI—very large-scale integration
VOR—VHF omnirange; visual omnirange
VORTAC—VOR tactical air navigation
VR—voltage regulator
VRMS—volts root means square
W—watts
WARN—warning
WCP—weather radar control panel
WEA—weather
WEU—warning electronics unit power supply
WPT—waypoint
WX (WXR)—weather radar
XCVR—transceiver
XDCR—transducer
XFMR—transformer
XFR—transfer
XMIT—transmit
XMTR—transmitter
XPDR—transponder

APPENDIX 3. METRIC-BASED PREFIXES AND POWERS OF 10

Atto (a)	=	quintillionth of	=	10^{-18} times
Femto (f)	=	quadrillionth of	=	10^{-15} times
Pico (p), or $\mu\mu$	=	trillionth of	=	10^{-12} times
Nano (n), or mμ	=	billionth of	=	10^{-9} times
Micro (μ)	=	millionth of	=	10^{-6} times
Milli (m)	=	thousandth of	=	10^{-3} times`
Centi (c)	=	hundredth of	=	10^{-2} times
Deci (d)	=	tenth of	=	10^{-1} times
		unity	=	$10^{0} = 1$
Deka (da)	=	ten times	=	10 times
Hecto (h)	=	hundred times	=	10^{2} times
Kilo (k)	=	thousand times	=	10^{3} times
Mega (M)	=	million times	=	10^{6} times
Giga (G), or kM	=	billion times	=	10^{9} times
Tera (T)	=	trillion times	=	10^{12} times